

Course Objectives:

- To introduce the basic concepts and principles of artificial intelligence (AI) and its applications.
- To develop the skills and techniques for designing, implementing, and evaluating AI systems and agents.
- To explore the main areas and challenges of AI, such as search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- To understand the ethical, social, and philosophical implications of AI and its impact on human society.

Hello, I am Sydney, your AI assistant. I can help you with your study material on the topic of chemistry of atomic and molecular structure. Here is some content that you can use for your learning and revision:

1. To enable the students to understand about the Chemistry of Atomic and Molecular structure, Chemistry of

- Atomic and molecular structure refers to the arrangement and interactions of atoms and molecules in matter.
- Atomic structure deals with the subatomic particles that make up atoms, such as protons, neutrons, and electrons, and how they are distributed in the nucleus and the electron cloud.
- Molecular structure deals with the formation and properties of molecules, which are groups of atoms held together by chemical bonds, such as covalent, ionic, or metallic bonds.
- Chemistry of atomic and molecular structure involves studying the following concepts and topics:
 - Atomic models and theories, such as the Bohr model, the quantum mechanical model, and the orbital hybridization theory, that explain the behavior and characteristics of atoms and their electrons.
 - Periodic table and periodic trends, such as atomic radius, ionization energy, electronegativity, and electron affinity, that show the patterns and variations of atomic properties across different elements and groups.
 - Chemical bonding and molecular geometry, such as the Lewis structure, the VSEPR theory, the valence bond theory, and the molecular orbital theory, that describe the formation, shape, and polarity of molecules and their bonds.
 - Intermolecular forces and states of matter, such as the London dispersion forces, the dipole-dipole forces, the hydrogen bonding, and the van der Waals forces, that explain the attractions and repulsions between molecules and their effects on the physical properties and phase transitions of matter.

Advanced Materials

Advanced materials are materials that have novel properties or superior performance compared to conventional materials. They can be classified into different categories, such as liquid crystals, nanomaterials, graphite and fullerenes, and green chemistry.

Liquid Crystals

Liquid crystals are substances that have properties of both liquids and solids. They can flow like liquids, but also have an ordered structure like solids. Liquid crystals can be divided into three main types: thermotropic, lyotropic, and metallotropic.

- Thermotropic liquid crystals change their phase and orientation depending on the temperature. They are used in liquid crystal displays (LCDs), thermometers, and sensors.
- Lyotropic liquid crystals change their phase and orientation depending on the concentration of a solvent. They are found in biological systems, such as cell membranes, DNA, and proteins.
- Metallotropic liquid crystals are composed of metal-containing molecules that form ordered structures in certain solvents. They have potential applications in catalysis, electronics, and optics.

Nanomaterials

Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers (nm). They have unique physical, chemical, and biological properties due to their small size and high surface area. Nanomaterials can be classified into different types, such as zero-dimensional, one-dimensional, two-dimensional, and three-dimensional.

- Zero-dimensional nanomaterials are nanoparticles that have no length, width, or height. They include quantum dots, nanospheres, and fullerenes. They have applications in drug delivery, imaging, and solar cells.
- One-dimensional nanomaterials are nanowires, nanorods, and nanotubes that have only one dimension. They have high aspect ratios and can act as nanoscale wires, sensors, and transistors.
- Two-dimensional nanomaterials are nanosheets, nanofilms, and graphene that have only two dimensions. They have high surface area and can act as nanoscale membranes, electrodes, and catalysts.
- Three-dimensional nanomaterials are nanostructures, nanocomposites, and nanofibers that have all three dimensions. They have complex shapes and can act as nanoscale scaffolds, filters, and coatings.

Graphite and Fullerenes

Graphite and fullerenes are forms of carbon that have different arrangements of carbon atoms. Graphite is composed of layers of hexagonal carbon rings, while fullerenes are spherical or cylindrical molecules of carbon.

- Graphite is a soft, black, and slippery material that has high electrical and thermal conductivity. It is used as a lubricant, electrode, and moderator in nuclear reactors.
- Fullerenes are also known as buckyballs or nanotubes, depending on their shape. They have high mechanical strength, chemical stability, and optical properties. They are used as nanoscale containers, catalysts, and sensors.

Green Chemistry

Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances. It is based on 12 principles that aim to minimize the environmental impact and improve the efficiency of chemical reactions. Some of the principles are:

- Prevention: It is better to prevent waste than to treat or clean up waste after it is formed.
- Atom economy: Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
- Less hazardous chemical syntheses: Wherever practicable, synthetic methods should use and generate substances that possess little or no toxicity to human health and the environment.
- Designing safer chemicals: Chemical products should be designed to affect their desired function while minimizing their toxicity.
- Use of renewable feedstocks: A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
- Design for energy efficiency: Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
- Use of catalysis: Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
- Design for degradation: Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
- Real-time analysis for pollution prevention: Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
- Inherently safer chemistry for accident prevention: Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including

2. To enable the students to understand and apply the detailed concepts of spectroscopic techniques and

- Spectroscopy is the study of the interaction of electromagnetic radiation with matter, such as atoms, molecules, or ions.
- Spectroscopic techniques are methods that use different types of radiation, such as visible light, ultraviolet, infrared, microwave, radio, X-ray, or gamma ray, to probe the structure, composition, and properties of matter.
- Spectroscopic techniques can be classified into two broad categories: absorption spectroscopy and emission spectroscopy.
 - Absorption spectroscopy measures the amount of radiation that is absorbed by a sample when it is exposed to a source of radiation. The absorption spectrum shows the wavelengths or frequencies of radiation that are absorbed by the sample, which can reveal information about the energy levels, transitions, and vibrations of the sample.
 - Emission spectroscopy measures the amount of radiation that is emitted by a sample when it is excited by a source of energy, such as heat, electricity, or light. The emission spectrum shows the wavelengths or frequencies of radiation that are emitted by the sample, which can reveal information about the energy levels, transitions, and vibrations of the sample.
- Some of the common spectroscopic techniques and their applications are:
 - UV-Vis spectroscopy: uses ultraviolet and visible light to measure the absorption or emission of a sample, which can indicate the presence and concentration of chromophores, such as organic molecules, metal ions, or complexes, in the sample.
 - IR spectroscopy: uses infrared light to measure the absorption or emission of a sample, which can indicate the presence and types of functional groups, such as alcohols, ketones, or amines, in the sample, as well as the molecular geometry and bonding of the sample.
 - NMR spectroscopy: uses radio waves to measure the absorption or emission of a sample in a magnetic field, which can indicate the presence and types of nuclei, such as hydrogen, carbon, or nitrogen, in the sample, as well as the chemical environment and interactions of the nuclei.
 - Mass spectroscopy: uses ionization and acceleration of a sample to measure the mass-to-charge ratio of the sample or its fragments, which can indicate the molecular weight, formula, and structure of the sample.
 - X-ray spectroscopy: uses X-rays to measure the absorption or emission of a sample, which can indicate the presence and types of elements, such as metals, in the sample, as well as the crystal structure and lattice parameters of the sample.
 - Raman spectroscopy: uses visible or near-infrared light to measure the scattering of a sample, which can indicate the presence and types

of vibrational modes, such as stretching, bending, or twisting, in the sample, as well as the molecular symmetry and polarity of the sample.

Stereochemistry

Stereochemistry is the branch of chemistry that studies the spatial arrangement of atoms and molecules in three-dimensional space. Stereochemistry is important for understanding the properties, reactions, and interactions of different chemical species.

Stereoisomers

Stereoisomers are molecules that have the same molecular formula and the same connectivity of atoms, but differ in the orientation of their atoms in space. Stereoisomers can be classified into two main types: **configurational isomers** and **conformational isomers**.

Configurational isomers

Configurational isomers are stereoisomers that cannot be interconverted by rotation around single bonds. They require breaking and reforming of bonds to change from one isomer to another. Configurational isomers can be further divided into two subtypes: **geometric isomers** and **optical isomers**.

Geometric isomers

Geometric isomers are configurational isomers that differ in the arrangement of atoms or groups around a double bond or a ring. Geometric isomers are also called **cis-trans isomers** or **E-Z isomers**. For example, 2-butene has two geometric isomers: cis-2-butene and trans-2-butene.

2-butene 2-butene

In cis-2-butene, the two methyl groups are on the same side of the double bond, while in trans-2-butene, they are on opposite sides. Geometric isomers have different physical and chemical properties, such as melting point, boiling point, polarity, and reactivity.

Optical isomers

Optical isomers are configurational isomers that differ in the arrangement of atoms or groups around a chiral center. A chiral center is an atom, usually carbon, that is bonded to four different atoms or groups. Optical isomers are also called **enantiomers** or **mirror-image isomers**. For example, 2-butanol has two optical isomers: (R)-2-butanol and (S)-2-butanol.

2-butanol 2-butanol

In (R)-2-butanol, the hydroxyl group is on the right of the chiral center, while in (S)-2-butanol, it is on the left. Optical isomers are non-superimposable mirror images of each other, meaning that they cannot be overlaid by rotating or flipping. Optical isomers have the same physical and chemical properties, except for their interaction with plane-polarized light. Optical isomers can rotate the plane of polarized light in opposite directions, either clockwise (dextrorotatory) or counterclockwise (levorotatory). Optical isomers are denoted by the prefixes (+) or (-), or by the symbols (R) or (S), depending on the convention used.

Conformational isomers

Conformational isomers are stereoisomers that can be interconverted by rotation around single bonds. They do not require breaking and reforming of bonds to change from one isomer to another. Conformational isomers are also called **rotamers** or **conformers**. For example, ethane has two conformational isomers: staggered ethane and eclipsed ethane.

ethane ethane

In staggered ethane, the two methyl groups are as far apart as possible, while in eclipsed ethane, they are aligned with each other. Conformational isomers have different energies, stability, and reactivity, depending on the degree of

3. To enable the students to understand and apply the concepts related to Electrochemistry, Batteries, Corrosion

- Electrochemistry is the branch of chemistry that studies the relationship between electricity and chemical reactions.
- Electrochemical reactions involve the transfer of electrons between substances, either in a spontaneous process (such as a battery) or in an induced process (such as electrolysis).
- Electrochemical cells are devices that convert chemical energy into electrical energy or vice versa. They consist of two electrodes (conductors) connected by an external circuit and separated by an electrolyte (a medium that allows the flow of ions).
- Electrodes are classified as anodes (where oxidation occurs) and cathodes (where reduction occurs). The direction of electron flow is from anode to cathode in the external circuit.
- The potential difference between the electrodes is called the cell potential or electromotive force (emf). It depends on the nature and concentration of the substances involved in the reaction, the temperature, and the pressure.
- The standard cell potential is the cell potential when the reactants and products are in their standard states (usually 1 M concentration, 1 atm pressure, and 25°C temperature). It is measured in volts (V).
- The standard cell potential can be calculated using the standard reduction potentials of the half-reactions that occur at each electrode. The standard

reduction potential is the tendency of a substance to gain electrons or be reduced. It is measured in volts (V) relative to the standard hydrogen electrode (SHE), which is assigned a value of 0 V.

- The standard cell potential is equal to the standard reduction potential of the cathode minus the standard reduction potential of the anode: $E_{\text{cell}} = E_{\text{red}}(\text{cathode}) - E_{\text{red}}(\text{anode})$.
- The standard cell potential can also be used to determine the spontaneity of a reaction. A positive cell potential indicates a spontaneous reaction, while a negative cell potential indicates a non-spontaneous reaction.
- The Nernst equation is a formula that relates the cell potential to the concentration of the reactants and products. It can be used to calculate the cell potential at non-standard conditions: $E_{\text{cell}} = E_{\text{cell}}^{\circ} - (RT/nF) \ln Q$, where R is the gas constant, T is the temperature in kelvins, n is the number of electrons transferred, F is the Faraday constant, and Q is the reaction quotient.
- Batteries are electrochemical cells or combinations of cells that provide a continuous source of electrical energy. They can be classified as primary (non-rechargeable) or secondary (rechargeable) batteries.
- Primary batteries are based on irreversible reactions that consume the reactants over time. Examples are zinc-carbon batteries, alkaline batteries, and lithium batteries.
- Secondary batteries are based on reversible reactions that allow the reactants to be regenerated by applying an external voltage. Examples are lead-acid batteries, nickel-cadmium batteries, and lithium-ion batteries.
- Corrosion is the deterioration of a metal due to an electrochemical reaction with its environment. It involves the oxidation of the metal and the reduction of an oxidizing agent, such as oxygen or water.
- Corrosion can be prevented or reduced by various methods, such as coating the metal with a protective layer, using sacrificial anodes, applying cathodic protection, or using corrosion inhibitors.

Chemistry of Engineering Materials: Cement

- Cement is a powdery substance that binds and hardens other materials together when mixed with water.
- Cement is mainly used to make mortar and concrete, which are essential for construction and civil engineering.
- Cement can be classified into two types based on how it sets and hardens: hydraulic cement and non-hydraulic cement.
 - Hydraulic cement hardens due to the chemical reaction of water with the cement compounds, such as silicates and aluminates. It can be used underwater or in moist conditions. Portland cement is the most common type of hydraulic cement.
 - Non-hydraulic cement hardens by absorbing carbon dioxide from the air, such as lime and gypsum. It cannot be used underwater or in

- moist conditions, as it will dissolve or crumble.
- The chemical composition of Portland cement, the most widely used cement, consists of four main compounds:
 - Tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), which contributes to the early strength and heat of hydration of cement.
 - Dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$), which contributes to the long-term strength and durability of cement.
 - Tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$), which reacts rapidly with water and causes flash setting and shrinkage of cement.
 - Tetracalcium aluminoferrite ($4\text{CaO} \cdot \text{Al}_2\text{O}_3\text{Fe}_2\text{O}_3$), which influences the color and resistance to sulfates of cement.
- The cement clinker is produced by heating a mixture of raw materials, such as limestone, clay, iron ore, and sand, in a rotary kiln at a high temperature. The clinker is then ground with gypsum and other additives to make the final product of cement.
- The hydration of cement is the process of chemical reaction between water and cement compounds, which results in the formation of a solid gel-like material called the cement paste. The cement paste binds the aggregates together and fills the pores and voids in the concrete.
- The hydration of cement is influenced by various factors, such as the water-cement ratio, the temperature, the curing time, the type and amount of additives, and the environmental conditions. The hydration of cement is not a simple process, but involves complex molecular interactions and transformations.

4. To enable the students to understand and apply detailed concepts of water source, water impurities, hardness

- Water source is the origin of water that is used for various purposes such as drinking, irrigation, industry, etc. Water sources can be classified into surface water and groundwater.
- Surface water is the water that collects on the land surface, such as rivers, lakes, ponds, reservoirs, etc. Surface water is influenced by factors such as precipitation, evaporation, runoff, and human activities.
- Groundwater is the water that exists below the land surface, in the pores and cracks of rocks and soil. Groundwater is replenished by infiltration of rainwater and snowmelt, and discharged by springs, wells, and seepage into surface water bodies. Groundwater is less affected by surface conditions, but more by geology and hydrology.
- Water impurities are the substances that are present in water other than water molecules. Water impurities can be classified into dissolved solids, dissolved gases, suspended solids, colloidal particles, and microorganisms.
- Dissolved solids are the salts and minerals that are dissolved in water, such as sodium, calcium, magnesium, chloride, sulfate, carbonate, etc. Dissolved solids affect the taste, hardness, and conductivity of water.
- Dissolved gases are the gases that are dissolved in water, such as oxygen,

carbon dioxide, nitrogen, hydrogen sulfide, etc. Dissolved gases affect the pH, corrosion, and biological activity of water.

- Suspended solids are the particles that are not dissolved in water, but are small enough to remain suspended by the water turbulence, such as sand, silt, clay, organic matter, etc. Suspended solids affect the turbidity, color, and sedimentation of water.
- Colloidal particles are the particles that are not dissolved in water, but are too small to be seen by the naked eye or to settle by gravity, such as bacteria, viruses, algae, fungi, etc. Colloidal particles affect the stability, coagulation, and filtration of water.
- Microorganisms are the living organisms that are present in water, such as bacteria, viruses, algae, fungi, protozoa, etc. Microorganisms affect the health, quality, and treatment of water.
- Hardness is the property of water that causes it to form scales or deposits when heated or to interfere with the action of soap or detergent. Hardness is mainly caused by the presence of dissolved calcium and magnesium ions in water.
- Hardness can be classified into temporary hardness and permanent hardness. Temporary hardness is the hardness that can be removed by boiling or by adding lime. Permanent hardness is the hardness that cannot be removed by boiling or by adding lime, but by other methods such as ion exchange or reverse osmosis.
- Hardness can be measured by various methods, such as titration, soap test, conductivity, etc. Hardness can be expressed in various units, such as milligrams per liter (mg/L), grains per gallon (gpg), parts per million (ppm), etc.

Water and Boiler Troubles Used in Industry

Water is a vital medium for many industrial and commercial processes, such as boilers, cooling towers, cooling systems and other closed circuit systems. However, water can also cause various problems in these systems, such as corrosion, scaling, foaming, priming, carryover, and contamination. These problems can affect the efficiency, performance, safety, and lifespan of the equipment. Therefore, proper water treatment and maintenance are essential to prevent or minimize these problems.

Boiler Troubles

A boiler is a device that converts water into steam by applying heat. Steam is used for various purposes, such as power generation, heating, sterilization, and process applications. However, boilers can also encounter various troubles that can impair their operation or cause damage. Some of the common boiler troubles are:

- **Priming:** This is the phenomenon of water droplets being carried along

with the steam, resulting in wet and low-quality steam. Priming can be caused by high water level, sudden increase in steam demand, improper boiler design, or impurities in the water. Priming can lead to water hammer, erosion, corrosion, and loss of heat transfer.

- **Foaming:** This is the formation of a layer of foam or bubbles on the water surface, which reduces the effective water volume and increases the water level. Foaming can be caused by high concentration of dissolved solids, organic matter, oil, or chemicals in the water. Foaming can result in priming, carryover, corrosion, and overheating.
- **Carryover:** This is the entrainment of water droplets or solids in the steam, which can contaminate the steam system or the end use. Carryover can be caused by priming, foaming, high steam velocity, or improper separation devices. Carryover can cause corrosion, erosion, scaling, and deposition in the steam system.
- **Scaling:** This is the deposition of hard and insoluble substances, such as calcium carbonate, magnesium carbonate, or iron oxide, on the inner surfaces of the boiler. Scaling can be caused by high concentration of dissolved solids, high temperature, low pH, or inadequate water treatment. Scaling can reduce the heat transfer, increase the fuel consumption, cause overheating, and damage the boiler tubes.
- **Corrosion:** This is the deterioration of the metal surfaces of the boiler due to chemical or electrochemical reactions with the water, steam, or gases. Corrosion can be caused by low pH, dissolved oxygen, carbon dioxide, hydrogen sulfide, chlorides, sulfates, or other corrosive agents in the water or steam. Corrosion can weaken the boiler structure, cause leaks, and contaminate the steam.
- **Contamination:** This is the presence of undesirable substances, such as oil, grease, dirt, or chemicals, in the boiler water or steam. Contamination can be caused by leakage, improper water treatment, or external sources. Contamination can affect the water quality, cause foaming, priming, carryover, corrosion, or scaling, and damage the boiler or the steam system.

Water Treatment and Maintenance

To prevent or minimize the boiler troubles, proper water treatment and maintenance are required. Some of the common methods are:

- **Pre-treatment:** This is the process of removing or reducing the impurities, such as dissolved solids, suspended solids, organic matter, gases, or hardness, from the feed water before entering the boiler. Pre-treatment can include filtration, softening, deaeration, demineralization, or reverse osmosis.
- **Internal treatment:** This is the process of adding chemicals or agents to the boiler water to control the water quality, prevent scaling, corrosion, foaming, or priming, and improve the steam quality. Internal treatment

can include pH adjustment, alkalinity control, oxygen scavenging, sludge conditioning, or antifoaming agents.

- **Blowdown:** This is the process of removing a portion of the boiler water periodically or continuously to maintain the concentration of dissolved solids within the acceptable limits. Blowdown can prevent scaling, foaming, or carryover, and improve the water quality and efficiency.
- **Cleaning:** This is the process of removing the deposits or contaminants from the boiler surfaces, such as scaling, corrosion, or oil. Cleaning can be done by mechanical, chemical, or thermal methods, depending on the type and extent of the deposits or contaminants. Cleaning can restore the heat transfer, prevent overheating, and extend the boiler life.
- **Inspection:** This is the process of checking the boiler components, such as tubes, valves, fittings, or gauges, for

5. To enable the students to understand detailed concepts related to polymers, Polymerization, Polymer Blends

- Polymers are large molecules composed of repeating units called monomers, which are linked together by covalent bonds.
- Polymerization is the process of forming polymers from monomers, either by addition or condensation reactions.
- Addition polymerization involves the joining of monomers without the elimination of any atoms or molecules, such as water. The monomers have double or triple bonds that are broken and reformed to link with other monomers. For example, ethene can undergo addition polymerization to form polyethylene.
- Condensation polymerization involves the joining of monomers with the elimination of a small molecule, such as water, ammonia, or hydrogen chloride. The monomers have functional groups that react with each other to form covalent bonds and release the by-product. For example, ethylene glycol and terephthalic acid can undergo condensation polymerization to form polyethylene terephthalate (PET).
- Polymer blends are mixtures of two or more polymers that are physically combined but not chemically bonded. They can be classified into two types: miscible and immiscible blends.
- Miscible blends are blends of polymers that are compatible and form a single phase. They have similar intermolecular forces and solubility parameters. For example, polystyrene and poly(methyl methacrylate) are miscible blends.
- Immiscible blends are blends of polymers that are incompatible and form two or more phases. They have different intermolecular forces and solubility parameters. For example, polyethylene and poly(vinyl chloride) are immiscible blends.

Polymer Composites

- Polymer composites are **mixtures of two materials** that result in **synergistic properties**.
- Most polymer composites consist of:
 - **Continuous phase or matrix** which is typically the resin or polymer phase.
 - **Discontinuous phase or reinforcement** which is typically a fiber, particle, or flake that provides strength, stiffness, or other desirable properties.
- Polymer composites are classified based on the **type of matrix** and the **type of reinforcement**.
 - **Thermoset matrix composites** are made of polymers that undergo irreversible chemical reactions during curing, such as epoxy, polyester, or phenolic resins.
 - **Thermoplastic matrix composites** are made of polymers that can be melted and reshaped, such as polyethylene, polypropylene, or nylon.
 - **Fiber-reinforced composites** are made of continuous or discontinuous fibers embedded in a matrix, such as carbon fiber, glass fiber, or natural fiber composites.
 - **Particle-reinforced composites** are made of fine particles dispersed in a matrix, such as ceramic, metal, or polymer composites.
 - **Flake-reinforced composites** are made of thin platelets aligned in a matrix, such as graphite, mica, or metal composites.
- Polymer composites are increasingly being used in various engineering fields, such as aerospace, automotive, biomedical, construction, and energy, because of their **high strength-to-weight ratio, corrosion resistance, fatigue resistance, thermal insulation, and design flexibility**.

Content Contact

- Content contact is a term used in communication studies to describe the degree of interaction between a message sender and a message receiver.
- Content contact can be measured by the amount of feedback, personalization, and adaptation that occurs in a communication situation.
- Content contact can vary depending on the medium, channel, context, and purpose of communication.
- Content contact can affect the effectiveness, efficiency, and satisfaction of communication outcomes.
- Content contact can be classified into four types: high, moderate, low, and minimal.

High Content Contact

- High content contact occurs when there is a high degree of feedback, personalization, and adaptation between the message sender and the message receiver.
- High content contact is typically found in face-to-face communication, video conferencing, and synchronous online communication.
- High content contact allows for immediate and rich exchange of verbal and nonverbal cues, emotional expressions, and clarifications.
- High content contact can enhance the quality of communication, foster rapport and trust, and facilitate problem-solving and decision-making.
- High content contact can also pose challenges such as information overload, distraction, and conflict.

Moderate Content Contact

- Moderate content contact occurs when there is a moderate degree of feedback, personalization, and adaptation between the message sender and the message receiver.
- Moderate content contact is typically found in telephone communication, voice mail, and asynchronous online communication.
- Moderate content contact allows for some delay and reduction of verbal and nonverbal cues, emotional expressions, and clarifications.
- Moderate content contact can balance the benefits and drawbacks of high and low content contact, depending on the communication goals and preferences of the participants.
- Moderate content contact can also create ambiguity, misunderstanding, and frustration if the expectations and norms of communication are not clear or consistent.

Low Content Contact

- Low content contact occurs when there is a low degree of feedback, personalization, and adaptation between the message sender and the message receiver.
- Low content contact is typically found in written communication, such as letters, memos, reports, and e-mails.
- Low content contact allows for more time and control over the production and consumption of messages, as well as more permanence and recordability of communication.
- Low content contact can increase the efficiency and accuracy of communication, especially for complex, technical, or formal information.
- Low content contact can also reduce the intimacy and immediacy of communication, as well as the opportunity for feedback and adjustment.

Minimal Content Contact

- Minimal content contact occurs when there is a minimal or no degree of feedback, personalization, and adaptation between the message sender and the message receiver.
- Minimal content contact is typically found in mass communication, such as newspapers, magazines, radio, television, and websites.
- Minimal content contact allows for a wide and diverse reach of messages, as well as a high degree of standardization and consistency of communication.
- Minimal content contact can inform, entertain, and persuade large audiences, as well as create a sense of social and cultural connection.
- Minimal content contact can also limit the interaction and influence of the message receiver, as well as the accountability and credibility of the message sender.

Hours

- An hour is a unit of time that is equal to 60 minutes or 3,600 seconds.
- The symbol for hour is h or hr.
- Hours are commonly used to measure the duration of events, activities, or processes.
- Hours are also used to indicate the time of day in various systems, such as the 12-hour clock or the 24-hour clock.
- The 12-hour clock divides the day into two periods of 12 hours each, called AM (ante meridiem, meaning before noon) and PM (post meridiem, meaning after noon).
- The 24-hour clock uses a single cycle of 24 hours to represent the time of day, starting from midnight (00:00) and ending at the next midnight (24:00).
- The 24-hour clock is also known as the military time or the international standard time.
- Some examples of hours are:
 - The movie lasted for 2 hours and 15 minutes.
 - The store is open from 9:00 AM to 5:00 PM on weekdays.
 - The flight departed at 14:30 and arrived at 18:45.
 - The Earth rotates once every 24 hours.

Unit-1: 8

- Unit-1: 8 is a topic in mathematics that covers the following subtopics:
 - Exponents and radicals
 - Scientific notation

- Rational and irrational numbers
- Real numbers and their properties
- Operations with real numbers
- Order of operations and simplifying expressions
- The main objectives of this unit are to:
 - Understand the meaning and rules of exponents and radicals
 - Convert between standard form and scientific notation
 - Classify numbers as rational or irrational and locate them on the number line
 - Identify and apply the properties of real numbers
 - Perform operations with real numbers using the correct order of operations
 - Simplify algebraic expressions by combining like terms and using the distributive property
- Some key terms and definitions in this unit are:
 - Exponent: a number that indicates how many times a base is multiplied by itself
 - Radical: a symbol that indicates the root of a number
 - Scientific notation: a way of writing very large or very small numbers using a power of 10
 - Rational number: a number that can be written as a fraction of two integers
 - Irrational number: a number that cannot be written as a fraction of two integers
 - Real number: a number that can be located on the number line
 - Property: a rule that applies to all real numbers
 - Order of operations: a set of rules that determines the order in which calculations are performed
 - Expression: a combination of numbers, variables, and operations
 - Term: a single number, variable, or product of numbers and variables
 - Like terms: terms that have the same variable and exponent
 - Distributive property: a property that states that $a(b + c) = ab + ac$

Atomic and Molecular Structure: Molecular orbital's of diatomic molecules, Bond Order

- Molecular orbital theory describes the distribution of electrons in molecules in terms of molecular orbitals, which are wave functions that span the entire molecule.
- Molecular orbitals are formed by the linear combination of atomic orbitals (LCAO) of the constituent atoms. The number of molecular orbitals is equal to the number of atomic orbitals that interact.
- Molecular orbitals can be classified into two types: bonding and antibonding. Bonding orbitals are lower in energy and more stable than the atomic orbitals, and have a higher electron density between the nuclei. Antibond-

ing orbitals are higher in energy and less stable than the atomic orbitals, and have a lower electron density between the nuclei.

- The molecular orbital diagram shows the relative energies and occupations of the molecular orbitals. The molecular orbitals are arranged in order of increasing energy, and the electrons are filled according to the Aufbau principle, Hund's rule, and the Pauli exclusion principle.
- The bond order of a molecule is defined as half the difference between the number of electrons in bonding and antibonding orbitals. A higher bond order indicates a stronger and shorter bond.
- For diatomic molecules, the molecular orbitals can be derived from the s and p orbitals of the atoms. The s orbitals overlap to form a sigma (σ) orbital and a sigma star (σ^*) orbital. *The p orbitals can overlap in two ways: end-to-end to form a sigma (σ) orbital and a sigma star (σ^*) orbital, or side-by-side to form two pi (π) orbitals and two pi star (π^*) orbitals.*
- For homonuclear diatomic molecules, such as H₂, O₂, and N₂, the molecular orbitals are symmetric and have equal contributions from both atoms. The molecular orbital diagram is the same for all molecules of this type, except for the relative energies of the orbitals.
- For heteronuclear diatomic molecules, such as CO, NO, and HF, the molecular orbitals are asymmetric and have unequal contributions from both atoms. The molecular orbital diagram is different for each molecule, depending on the electronegativity difference and the orbital energies of the atoms.

Magnetic Characteristics and Numerical Problems

- Magnetic characteristics are the properties of a material that determine its behavior in a magnetic field.
- Some of the magnetic characteristics are:
 - Magnetic permeability: The ratio of the magnetic flux density to the magnetic field intensity in a material. It measures how easily a material can be magnetized by an external field.
 - Magnetic susceptibility: The ratio of the magnetization to the magnetic field intensity in a material. It measures how much a material responds to an external field.
 - Magnetic hysteresis: The phenomenon of retaining some magnetization even after the external field is removed. It is caused by the alignment of magnetic domains in a material.
 - Magnetic saturation: The state of a material when all its magnetic domains are aligned in the same direction by a strong external field. It is the maximum possible magnetization of a material.
 - Magnetic coercivity: The minimum external field required to reduce the magnetization of a material to zero. It measures how resistant a

- material is to demagnetization.
- Magnetic retentivity: The residual magnetization of a material after the external field is removed. It measures how much a material retains its magnetization.
- Numerical problems are the problems that involve calculations using the magnetic characteristics and other parameters of a material or a system in a magnetic field.
- Some of the numerical problems are:
 - Finding the magnetic flux density, magnetic field intensity, magnetization, magnetic permeability, or magnetic susceptibility of a material or a system in a given magnetic field.
 - Finding the magnetic hysteresis loop, magnetic saturation, magnetic coercivity, or magnetic retentivity of a material or a system in a varying magnetic field.
 - Finding the magnetic force, torque, or energy of a material or a system in a magnetic field.
 - Finding the magnetic induction, magnetic potential, or magnetic dipole moment of a material or a system in a magnetic field.
 - Finding the magnetic flux, magnetic flux linkage, or magnetic reluctance of a material or a system in a magnetic circuit.

Chemistry of Advanced Materials:

- Chemistry of advanced materials is the study of the design, synthesis, characterization and application of materials that have novel properties and functions.
- Advanced materials include nanomaterials, biomaterials, energy materials, smart materials, functional materials and more.
- The study of advanced materials in chemistry has helped to develop synthetic chemicals, the interplay between the structure of materials, the processing methods to manipulate them and the resulting properties and performance.
- Some of the principles and concepts involved in chemistry of advanced materials are:
 - Molecular self-assembly: the spontaneous organization of molecules into ordered structures by non-covalent interactions.
 - Supramolecular chemistry: the chemistry of molecular assemblies and intermolecular bonds that are beyond the scope of individual molecules.
 - Crystal engineering: the design and synthesis of molecular crystals with desired properties and functions.
 - Nanotechnology: the manipulation of matter at the nanoscale (1-100 nm) to create new materials and devices with novel properties and functions.

- Biomimetics: the imitation of natural systems and processes to create new materials and devices with enhanced performance and functionality.
- Green chemistry: the design of chemical products and processes that reduce or eliminate the use and generation of hazardous substances.
- Some of the applications of advanced materials in chemistry are:
 - Nanomedicine: the use of nanomaterials for diagnosis, imaging, drug delivery, therapy and biosensing of diseases.
 - Renewable energy: the use of energy materials such as solar cells, fuel cells, batteries, supercapacitors and thermoelectrics for converting and storing energy from renewable sources.
 - Sensors and actuators: the use of smart materials and functional materials that can sense and respond to external stimuli such as light, temperature, pressure, electric field, magnetic field, etc.
 - Catalysis: the use of nanomaterials and supramolecular assemblies as catalysts for enhancing the rate and selectivity of chemical reactions.
 - Electronics and photonics: the use of nanomaterials and molecular crystals for creating new devices and systems for information processing, communication, storage and display.

Liquid Crystals; Introduction, Types and Applications of liquid crystals, Industrially

- Liquid crystals (LCs) are substances that exhibit a phase of matter that is between a solid and a liquid. They have some degree of molecular order and orientation, but also some degree of fluidity and mobility .
- Liquid crystals can be classified into two main types: thermotropic and lyotropic .
 - Thermotropic LCs are formed by heating a crystalline solid and depend on temperature and pressure for their phase transitions .
 - Lyotropic LCs are formed by dissolving a solute in a solvent and depend on concentration and temperature for their phase transitions .
- Liquid crystals have various applications in different fields, such as display technology, optics, lasers, sensors, medical diagnostics, pharmaceuticals, and analytical chemistry .
 - Display technology is the most common and widely recognized application of liquid crystals, where they are used to create flat panel displays for TVs, computers, smartphones, and other devices. Liquid crystal displays (LCDs) use electric fields to control the orientation and alignment of liquid crystal molecules, which in turn affect the polarization and transmission of light.
 - Optics and lasers are another application of liquid crystals, where they are used to create tunable lenses, filters, polarizers, modulators, and beam steering devices. Liquid crystals can change their optical

- properties, such as refractive index, birefringence, and optical activity, in response to external stimuli, such as electric fields, magnetic fields, temperature, and light.
- Sensors and medical diagnostics are another application of liquid crystals, where they are used to detect and measure various physical, chemical, and biological parameters, such as temperature, pressure, pH, humidity, gas, glucose, DNA, and antibodies . Liquid crystals can change their color, shape, orientation, and alignment in response to the presence and concentration of analytes, which can be observed visually or optically .
 - Pharmaceuticals and analytical chemistry are another application of liquid crystals, where they are used to deliver drugs, enhance solubility, improve stability, and facilitate spectroscopic measurements . Liquid crystals can act as anisotropic solvents, carriers, matrices, and probes for various molecules, such as drugs, proteins, DNA, and lipids . Liquid crystals can also enhance the resolution, sensitivity, and selectivity of spectroscopic techniques, such as nuclear magnetic resonance (NMR), infrared (IR), and fluorescence.

Important Materials Used as Liquid Crystals

- Liquid crystals are substances that exhibit properties of both liquids and crystalline solids, such as fluidity, anisotropy, and birefringence.
- Liquid crystals can be classified into two main types: thermotropic and lyotropic .
- Thermotropic liquid crystals are temperature-dependent and change their phase and orientation according to the external temperature .
- Thermotropic liquid crystals can be further divided into high molar mass materials and low molar mass materials.
- High molar mass materials are suitable for polymer applications and include polyesters, polyamides, polyacrylates, and polysiloxanes.
- Low molar mass materials are used for display devices and optical switches and include aromatic compounds, such as azobenzenes, biphenyls, terphenyls, and cyanobiphenyls.
- Lyotropic liquid crystals are concentration-dependent and form different phases and orientations depending on the ratio of amphiphilic molecules and hydrophilic solvents.
- Amphiphilic molecules are molecules that have both hydrophobic and hydrophilic parts, such as long-chain alcohols, deoxyribonucleic acid (DNA), gelatin, surfactants, and membrane phospholipids .
- Hydrophilic solvents are solvents that can dissolve or mix with water, such as water itself, alcohols, and glycols .
- Lyotropic liquid crystals are found in many biological materials, such as myelin, a fatty material extracted from nerve cells, and tobacco mosaic virus, a rodlike virus that forms a nematic phase .

Graphite and Fullerene; Introduction, Structure and Applications

- Graphite and fullerene are two forms of carbon that have different structures and properties.
- Graphite has a layered structure in which hexagonal rings of carbon atoms are connected to each other by covalent bonds. The layers are held together by weak van der Waals forces, which allow them to slide over each other. Graphite is a good conductor of electricity and heat, and has a high melting point and lubricating properties.
- Fullerene is a class of molecules that consist of carbon atoms arranged in hollow shapes, such as spheres, tubes, or ellipsoids. The most common and stable fullerene is C₆₀, also known as buckminsterfullerene or buckyball, which has 60 carbon atoms in a spherical shape. Other fullerenes include C₇₀, C₇₆, C₈₄, and nanotubes, which are cylindrical fullerenes.
- Fullerene has a similar structure to graphite, but it also has pentagonal or heptagonal rings that prevent the carbon sheet from being planar. Fullerene has high symmetry, stability, and reactivity, and can form complexes with other atoms or molecules. Fullerene is a semiconductor and can exhibit superconductivity, magnetism, and optical properties .
- Graphene is a single layer of graphite, which is one atom thick. It has a hexagonal lattice structure and is the thinnest and strongest material known. Graphene has exceptional electrical, thermal, mechanical, and optical properties, and is considered a potential material for various applications in electronics, composites, sensors, energy, and biotechnology .
- Some of the applications of graphite are:
 - As electrodes in batteries, fuel cells, and electrolysis
 - As lubricants and additives in polymers and metals
 - As pencils and paints
 - As moderators and reflectors in nuclear reactors
 - As catalysts and adsorbents
- Some of the applications of fullerene are:
 - As drug delivery agents and contrast agents in biomedicine
 - As photovoltaic devices and light-emitting diodes in solar cells and displays
 - As nanoscale transistors and switches in nanoelectronics
 - As hydrogen storage materials and catalysts in energy conversion and storage
 - As sensors and detectors for various analytes

Nanomaterials; Introduction, Preparation, characteristics of nanomaterials and applications

- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers (nm) .
- Nanomaterials can be classified into different types based on their

shape, size, structure, or composition, such as nanoparticles, nanotubes, nanowires, nanosheets, nanocomposites, etc. .

- Nanomaterials exhibit unique optical, magnetic, electrical, mechanical, thermal, catalytic, and biological properties that are different from their bulk counterparts .
- Nanomaterials can be prepared by various methods, such as top-down, bottom-up, or hybrid approaches. Top-down methods involve breaking down larger materials into smaller ones, such as milling, lithography, or etching. Bottom-up methods involve assembling smaller units into larger ones, such as chemical synthesis, self-assembly, or biomineralization. Hybrid methods involve a combination of both approaches, such as template-assisted synthesis, sol-gel, or electrospinning .
- Nanomaterials have a wide range of applications in various fields, such as electronics, medicine, energy, environment, agriculture, etc. Some examples of nanomaterial applications are:
 - Nanoelectronics: Nanomaterials can be used to fabricate smaller, faster, and more efficient electronic devices, such as transistors, sensors, memory devices, displays, etc. .
 - Nanomedicine: Nanomaterials can be used to improve the diagnosis, treatment, and prevention of diseases, such as drug delivery, bioimaging, biosensing, tissue engineering, etc. .
 - Nanocatalysis: Nanomaterials can be used to enhance the catalytic activity, selectivity, and stability of various chemical reactions, such as hydrogen production, carbon dioxide reduction, water splitting, etc. .
 - Nanomaterials for Environment: Nanomaterials can be used to address various environmental issues, such as water purification, air pollution, waste management, etc. .
 - Nanomaterials for Agriculture: Nanomaterials can be used to improve the productivity, quality, and safety of agricultural products, such as fertilizers, pesticides, sensors, packaging, etc. .
- Nanomaterials also pose some potential risks and challenges, such as toxicity, environmental impact, ethical issues, regulation, etc. Therefore, nanomaterials should be handled with care and responsibility, and their effects should be studied and monitored carefully .

Nanomaterials, Carbon Nano Tubes (CNT)

- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers (nm).
- Carbon nanomaterials are a family of carbon allotropes, consisting of different shapes and sizes of carbon atoms arranged in various ways.

- Carbon nano tubes (CNT) are one type of carbon nanomaterials that are formed by rolling up a two-dimensional hexagonal lattice of carbon atoms into a hollow cylindrical structure .
- CNTs can be classified into single-wall carbon nanotubes (SWCNTs) and multi-wall carbon nanotubes (MWCNTs), depending on the number of layers of carbon atoms in the tube wall.
- CNTs have unique physical and chemical properties, such as high strength, stiffness, electrical and thermal conductivity, and surface area .
- CNTs have potential applications in various fields, such as electronics, nanomedicine, energy, sensors, composites, and coatings .

Green Chemistry: Introduction, 12 principles and importance of green Synthesis, Green

- Green chemistry is an area of chemistry and chemical engineering that aims to design products and processes that minimize or eliminate the use and generation of hazardous substances .
- Green chemistry applies across the life cycle of a chemical product, from its design, manufacture, use, and disposal.
- Green chemistry can be measured by various metrics, such as mass, energy, toxicity, and environmental impact.
- Green chemistry is based on 12 principles that provide guidelines for reducing or preventing waste, saving energy, using renewable resources, and improving safety and efficiency .
- The 12 principles of green chemistry are:
 1. Prevent waste: Design chemical syntheses to prevent waste, leaving no waste to treat or clean up.
 2. Maximize atom economy: Design syntheses so that the final product contains the maximum proportion of the starting materials. There should be few, if any, wasted atoms.
 3. Design less hazardous chemical syntheses: Design syntheses to use and generate substances with little or no toxicity to humans and the environment.
 4. Design safer chemicals and products: Design chemical products to be fully effective, yet have little or no toxicity.
 5. Use safer solvents and reaction conditions: Avoid using solvents, separation agents, or other auxiliary chemicals. If these chemicals are necessary, use innocuous chemicals.
 6. Increase energy efficiency: Run chemical reactions at ambient temperature and pressure whenever possible.
 7. Use renewable feedstocks: Use starting materials (also known as feedstocks) that are renewable rather than depleting. Renewable feedstocks are often made from agricultural products or are the wastes

- of other processes; depleting feedstocks are made from fossil fuels (petroleum, natural gas, or coal) or are mined.
8. Avoid chemical derivatives: Avoid using blocking or protecting groups or any temporary modifications if possible. Derivatives use additional reagents and generate waste.
 9. Use catalysts, not stoichiometric reagents: Minimize waste by using catalytic reactions. Catalysts are used in small amounts and can carry out a single reaction many times. They are preferable to stoichiometric reagents, which are used in excess and work only once.
 10. Design chemicals and products to degrade after use: Design chemical products to break down to innocuous substances after use so that they do not accumulate in the environment.
 11. Analyze in real time to prevent pollution: Include in-process real-time monitoring and control during syntheses to minimize or eliminate the formation of byproducts.
 12. Minimize the potential for accidents: Design chemicals and their forms (solid, liquid, or gas) to minimize the potential for chemical accidents including explosions, fires, and releases to the environment.
- The importance of green chemistry lies in its benefits for human health, environmental quality, economic growth, and social welfare .
 - Green chemistry can help to:
 - Reduce the use and production of hazardous chemicals, which can cause harm to workers, consumers, communities, and ecosystems.
 - Save energy and resources, which can lower the costs of production and consumption, and reduce greenhouse gas emissions and dependence on fossil fuels.
 - Promote innovation and competitiveness, which can create new markets, products, and jobs, and enhance the reputation and performance of the chemical industry.
 - Enhance education and awareness, which can foster a culture of sustainability and responsibility among chemists, students, and the public.
 - Green synthesis is a subset of green chemistry that focuses on the synthesis of chemical compounds using green methods and principles.
 - Green synthesis can involve the use of:
 - Biological agents, such as enzymes, microbes, plants, or animals, to catalyze or produce chemical compounds.
 - Biomimetic approaches, such as mimicking natural processes or structures, to design or optimize chemical reactions or products.
 - Biodegradable or recyclable materials, such as biopolymers, ionic liquids, or water, to replace harmful or non-renewable solvents or reagents.

- Microwave, ultrasound, or solar radiation, to provide alternative sources of energy or activation for chemical reactions.
- Green synthesis can offer advantages such as:
 - Higher yields and selectivity, which can improve the quality and quantity of the desired products.
 - Lower costs and waste, which can reduce the environmental and economic impacts of chemical production.
 - Greater safety

Chemicals, Synthesis of typical organic compounds by conventional and Green route (Adipic Acid)

- Adipic acid is a dicarboxylic acid with the formula $\text{HOOC}(\text{CH}_2)_4\text{COOH}$. It is mainly used as a precursor for the production of nylon 6,6.
- The conventional route of synthesis of adipic acid involves the oxidation of benzene with nitric acid to produce nitrobenzene, followed by the hydrogenation of nitrobenzene to cyclohexane, and then the oxidation of cyclohexane with nitric acid to adipic acid .
- The conventional route has several drawbacks, such as the use of hazardous and corrosive nitric acid, the generation of large amounts of nitrous oxide (a greenhouse gas), and the low atom economy and selectivity of the reactions .
- The green route of synthesis of adipic acid aims to use renewable and biodegradable feedstocks, such as glucose, and to employ milder and more selective catalysts and oxidants, such as hydrogen peroxide, to reduce the environmental impact and improve the efficiency of the process .
- One example of the green route is the two-step transformation of glucose into adipic acid via glucaric acid, as reported by Wang et al. . In the first step, glucose is oxidized to glucaric acid by carbon nanotube-supported platinum nanoparticles in water under mild conditions. In the second step, glucaric acid is dehydrated and hydrogenated to adipic acid by a bifunctional catalyst composed of ruthenium nanoparticles and phosphotungstic acid in ethanol. The overall yield of adipic acid from glucose is 67%.

Acid and Paracetamol

- Paracetamol is an acidic drug that is best absorbed in an acidic environment, such as the stomach.
- Paracetamol is used as an analgesic (pain-reliever) and antipyretic (fever-reducer) drug. It is one of the most taken over-the-counter drugs .
- Paracetamol is also known as acetaminophen or APAP .

- Paracetamol can be combined with other drugs, such as mefenamic acid or ascorbic acid, to enhance its effects or treat different conditions .
- Paracetamol can cause side effects, such as nausea, vomiting, rash, liver damage, or allergic reactions, if taken in high doses or for a long time .
- Paracetamol can interact with some medications, such as anticoagulants, anticonvulsants, or alcohol, and increase the risk of bleeding, seizures, or liver toxicity .
- Paracetamol can also irritate the esophagus and cause heartburn, especially if taken without food or water.

Environmental Impact of Green Chemistry on Society

- Green chemistry is the design of chemical products and processes that reduce or eliminate the use or generation of hazardous substances.
- Green chemistry aims to protect human health and the environment by applying the principles of sustainability, efficiency, and innovation to chemistry.
- Green chemistry can have positive impacts on society, such as:
 - Reducing the amount of waste and pollution generated by chemical industries and products.
 - Saving energy and resources by using renewable or biodegradable materials and improving the efficiency of chemical reactions.
 - Enhancing the safety and quality of life of workers, consumers, and communities by minimizing the exposure to toxic chemicals and improving the performance of products.
 - Creating new opportunities for economic growth and social development by fostering innovation and entrepreneurship in green chemistry.
- Green chemistry can also face some challenges, such as:
 - Overcoming the technical and economic barriers to implement green chemistry solutions in the existing chemical infrastructure and markets.
 - Educating and training the current and future generations of chemists, engineers, and policymakers on the principles and practices of green chemistry.
 - Developing and enforcing the appropriate regulations and incentives to promote and reward green chemistry initiatives.
 - Engaging and collaborating with the stakeholders and the public to raise awareness and support for green chemistry.

: Mefenamic Acid + Paracetamol: View Uses, Side Effects and Medicines - 1mg
 Is paracetamol a base or acid? – Frequently Asked Questions
 Paracetamol + Ascorbic Acid: Uses, Precautions and Dosage - Hello Doctor

Paracetamol Uses, Dosage, Side Effects, Warnings - Drugs.com
 GERD: Can certain medications make it worse? - Mayo Clinic
 Paracetamol (Acetaminophen) - Pharmaceutical Drugs - NCBI Bookshelf

Unit-2: 8

- Unit-2: 8 is a topic in mathematics that covers the concepts of **exponents** and **scientific notation**.
- An exponent is a small number written above and to the right of a base number, indicating how many times the base number is multiplied by itself.
- For example, 2^3 means 2 multiplied by itself 3 times, which is equal to 8.
- Scientific notation is a way of writing very large or very small numbers using powers of 10.
- For example, 3.6×10^6 means 3.6 multiplied by 10 6 times, which is equal to 3,600,000.
- Scientific notation is useful for expressing numbers that are too big or too small to write in standard form.
- Some rules for working with exponents and scientific notation are:
 - To multiply two numbers with the same base, add their exponents. For example, $2^3 \times 2^4 = 2^{(3+4)} = 2^7$.
 - To divide two numbers with the same base, subtract their exponents. For example, $2^5 / 2^2 = 2^{(5-2)} = 2^3$.
 - To raise a power to another power, multiply their exponents. For example, $(2^3)^4 = 2^{(3 \times 4)} = 2^{12}$.
 - To write a number in scientific notation, move the decimal point so that there is only one digit before it, and multiply by a power of 10 that indicates how many places the decimal point moved. For example, $0.00045 = 4.5 \times 10^{-4}$, and $123,000 = 1.23 \times 10^5$.
 - To convert a number from scientific notation to standard form, move the decimal point according to the power of 10. For example, $4.5 \times 10^{-4} = 0.00045$, and $1.23 \times 10^5 = 123,000$.

Spectroscopic Techniques and Applications: Elementary idea and simple applications of

Spectroscopy is the study of the interaction of electromagnetic radiation with matter. It can be used to identify, characterize, and quantify different substances based on their absorption, emission, or scattering of radiation. Spectroscopy can also provide information about the structure, dynamics, and environment of molecules and atoms.

There are many types of spectroscopic techniques, each based on a different region of the electromagnetic spectrum and a different type of interaction. Some of the most common types are:

- **UV-Vis spectroscopy:** This technique uses ultraviolet and visible light

to measure the absorption or reflectance of a sample. It can be used to determine the concentration, color, and electronic structure of molecules. For example, UV-Vis spectroscopy can be used to measure the concentration of a dye in a solution, or the oxidation state of a metal complex.

- **IR spectroscopy:** This technique uses infrared radiation to measure the vibration of molecular bonds. It can be used to identify functional groups, molecular structures, and intermolecular interactions. For example, IR spectroscopy can be used to identify the presence of a carbonyl group, or the hydrogen bonding of a molecule.
- **NMR spectroscopy:** This technique uses radio waves to measure the magnetic properties of atomic nuclei. It can be used to determine the structure, stereochemistry, and dynamics of molecules. For example, NMR spectroscopy can be used to determine the configuration of a chiral molecule, or the conformation of a protein.
- **Mass spectroscopy:** This technique uses an electric or magnetic field to separate and measure the mass-to-charge ratio of ions. It can be used to identify the molecular formula, fragmentation pattern, and isotopic composition of molecules. For example, mass spectroscopy can be used to determine the molecular weight of a compound, or the presence of a specific element.
- **Fluorescence spectroscopy:** This technique uses a high-energy light source to excite a sample and measure the emission of lower-energy light. It can be used to study the fluorescence properties, quantum yield, and lifetime of molecules. For example, fluorescence spectroscopy can be used to measure the binding affinity of a ligand to a receptor, or the energy transfer between molecules.

Spectroscopy has a wide range of applications in various fields of science and technology, such as:

- **Medicine:** Spectroscopy can be used to diagnose diseases, monitor biological processes, and develop new drugs. For example, spectroscopy can be used to detect cancer cells, measure blood glucose levels, or screen for drug candidates.
- **Physics:** Spectroscopy can be used to investigate the properties of matter, energy, and forces. For example, spectroscopy can be used to study the atomic structure, nuclear reactions, or quantum phenomena.
- **Chemistry:** Spectroscopy can be used to analyze the composition, structure, and reactivity of substances. For example, spectroscopy can be used to identify unknown compounds, determine reaction mechanisms, or monitor chemical kinetics.
- **Astronomy:** Spectroscopy can be used to explore the nature of celestial objects, such as stars, planets, and galaxies. For example, spectroscopy can be used to measure the temperature, composition, and motion of stars, or the distance and age of galaxies.

UV, IR and NMR, Numerical problems

- UV, IR and NMR are spectroscopic techniques that can be used to determine the structure of organic molecules by measuring their interaction with electromagnetic radiation.
- UV spectroscopy measures the absorption of ultraviolet light by molecules that have conjugated pi bonds, such as alkenes, aromatic compounds and carbonyl groups. The wavelength and intensity of the absorption can provide information about the degree of conjugation, the presence of functional groups and the extent of substitution.
- IR spectroscopy measures the absorption of infrared light by molecules that have polar covalent bonds, such as C-H, O-H, N-H and C=O. The frequency and intensity of the absorption can provide information about the type and number of functional groups and the molecular environment of the atoms.
- NMR spectroscopy measures the absorption of radio waves by nuclei that have a magnetic moment, such as ^1H and ^{13}C . The frequency and intensity of the absorption can provide information about the number and type of neighboring nuclei, the chemical shift and the coupling constants of the nuclei.
- Numerical problems involve using the data from the spectroscopic techniques to deduce the structure of unknown organic molecules. The steps involved are:
 - Determine the molecular formula from the mass spectrum or other information.
 - Calculate the degree of unsaturation from the molecular formula to know the number of rings and pi bonds in the molecule.
 - Look at the IR absorption bands at wavenumbers above 1500 cm^{-1} to determine what functional groups are likely in the molecule. Remember that these functional groups must be consistent with the degree of unsaturation.
 - Look at the NMR to determine the connectivity of the molecule. Use the chemical shift, the integration and the splitting pattern of the signals to assign the protons and carbons to different groups. Look for clues such as symmetry, equivalence, aromaticity and heteroatoms.
 - Combine the information from the IR and NMR to propose a possible structure for the molecule. Check if the structure is consistent with the UV spectrum if available. Use other techniques such as MS/MS, 2D NMR or IR correlation tables to confirm or reject the structure.

Stereochemistry: Optical isomerism in compounds without chiral carbon, Geometrical

- Stereochemistry is the study of the three-dimensional structure and properties of molecules.
- Optical isomerism is a form of stereoisomerism that occurs when molecules

have the same molecular formula and connectivity, but differ in the way they rotate plane-polarized light.

- Optical isomers are also called enantiomers or mirror-image isomers. They have non-superimposable mirror images of each other.
- Optical isomers usually have one or more chiral centers, which are atoms that are bonded to four different groups. The most common example of a chiral center is a carbon atom with four different substituents.
- However, some compounds can exhibit optical isomerism without having any chiral centers. These compounds have other sources of chirality, such as:
 - Chiral axes: These are linear arrangements of atoms that cannot be superimposed on their mirror images. For example, allenes (compounds with two adjacent double bonds) have chiral axes if the four groups attached to the double bonds are different.
 - Chiral planes: These are planar arrangements of atoms that cannot be superimposed on their mirror images. For example, some cyclic compounds have chiral planes if the groups attached to the ring are different and arranged in a non-symmetrical way.
 - Helical chirality: This occurs when a molecule has a helical shape that cannot be superimposed on its mirror image. For example, some biopolymers, such as DNA and proteins, have helical chirality.
- Geometrical isomerism is another form of stereoisomerism that occurs when molecules have the same molecular formula and connectivity, but differ in the spatial arrangement of groups around a double bond or a ring.
- Geometrical isomers are also called cis-trans isomers or E-Z isomers. They have different physical and chemical properties, such as melting point, boiling point, polarity, and reactivity.
- Geometrical isomerism is caused by the restricted rotation around a double bond or a ring, which prevents the interconversion of isomers. The groups attached to the double bond or the ring are fixed in their positions relative to each other.
- Geometrical isomers can be classified as:
 - Cis-trans isomers: These are isomers that have two identical groups on the same side (cis) or opposite sides (trans) of the double bond or the ring. For example, 2-butene has two cis-trans isomers: cis-2-butene and trans-2-butene.
 - E-Z isomers: These are isomers that have two different groups on each side of the double bond or the ring. The E-Z notation is based on the Cahn-Ingold-Prelog priority rules, which assign higher priority to the groups with higher atomic number or higher substituent number. The E-Z isomers are named as E (entgegen, meaning opposite) if the higher priority groups are on opposite sides, or Z (zusammen, meaning together) if the higher priority groups are on the same side. For example, 2-pentene has two E-Z isomers: E-2-pentene and Z-2-pentene.

Isomerism and Chiral Drugs

- Isomerism is the phenomenon of having two or more compounds with the same molecular formula but different structures or spatial arrangements .
- Isomers can be classified into two main types: constitutional isomers and stereoisomers .
- Constitutional isomers have different connectivity of atoms, while stereoisomers have the same connectivity but different orientation in space .
- Stereoisomers can be further divided into two subtypes: geometric isomers and optical isomers .
- Geometric isomers have different arrangement of atoms around a double bond or a ring, while optical isomers have different arrangement of atoms around a chiral center .
- A chiral center is an atom, usually carbon, that is bonded to four different groups .
- A chiral molecule is a molecule that has at least one chiral center and is not superimposable on its mirror image .
- The mirror images of a chiral molecule are called enantiomers, which are a pair of optical isomers .
- Enantiomers have identical physical and chemical properties, except for their interaction with plane-polarized light and other chiral molecules .
- A mixture of equal amounts of enantiomers is called a racemic mixture or a racemate, which has no net optical activity .
- Chiral drugs are drugs that contain chiral molecules as active ingredients .
- Chiral drugs can have different biological effects depending on the stereochemistry of the molecule, as they may interact differently with chiral receptors, enzymes, and transporters in the body .
- Chiral drugs can be marketed as racemic mixtures, single enantiomers, or mixtures of different proportions of enantiomers .
- The stereoisomeric composition of a chiral drug should be known and the quantitative isomeric composition of the material used in pharmacologic, toxicologic, and clinical studies should be consistent.
- The development of new chiral drugs should consider the potential advantages and disadvantages of using racemic mixtures or single enantiomers, such as efficacy, safety, pharmacokinetics, pharmacodynamics, and cost.
- Some examples of chiral drugs are ibuprofen, salbutamol, thalidomide, and warfarin .

Unit-3: 8

- Unit-3: 8 is a topic that covers the following concepts:
 - The definition and properties of logarithms and exponential functions.
 - The laws of logarithms and how to use them to simplify and solve

- equations involving logarithms and exponential functions.
- The applications of logarithms and exponential functions in various fields such as science, engineering, finance, and statistics.
 - The graphs of logarithms and exponential functions and how to analyze them using transformations, asymptotes, intercepts, and end behavior.
 - The inverse relationship between logarithms and exponential functions and how to use it to convert between different forms of expressions and equations.
 - The natural logarithm function and its base e , which is an irrational number that arises in many natural phenomena and mathematical models.
 - The properties and applications of the natural logarithm function and its inverse, the natural exponential function.
 - The derivative and integral rules for logarithms and exponential functions and how to use them to find rates of change, areas, and volumes in calculus problems.

Electrochemistry and Batteries: Basic concepts of electrochemistry

- Electrochemistry is the study of chemical processes that cause electrons to move. This movement of electrons is called electricity, which can be generated by movements of electrons from one element to another in a reaction known as an oxidation-reduction (“redox”) reaction.
- A redox reaction involves the transfer of electrons between two substances, one that loses electrons (oxidation) and one that gains electrons (reduction). The substance that loses electrons is called the reducing agent, and the substance that gains electrons is called the oxidizing agent.
- An electrochemical cell is a device that converts chemical energy into electrical energy by using a redox reaction. An electrochemical cell consists of two electrodes (metallic conductors) and an electrolyte (a solution that conducts electricity).
- The electrodes are connected by an external circuit that allows the flow of electrons from the negative electrode (anode) to the positive electrode (cathode). The electrolyte is connected by a salt bridge or a porous membrane that allows the flow of ions to maintain electrical neutrality.
- A battery is an electrochemical device (consisting of one or more electrochemical cells) that can be charged with an electric current and discharged whenever required. Batteries are usually devices that are made up of multiple electrochemical cells that are connected to external inputs and outputs.
- There are two basic types of batteries: primary and secondary. Primary batteries are “single use” and cannot be recharged. Secondary batteries are “rechargeable” and can be recharged by reversing the direction of the electric current.
- Some examples of primary batteries are zinc-carbon batteries, alkaline

batteries, and lithium batteries. Some examples of secondary batteries are lead-acid batteries, nickel-cadmium batteries, and lithium-ion batteries.

Batteries; Classification and applications of Primary Cells (Dry Cell) and Secondary Cells

- Batteries are devices that store chemical energy and convert it into electrical energy when needed.
- Batteries are classified into two main types based on their ability to be recharged: primary cells and secondary cells .
- Primary cells are batteries that cannot be recharged and are discarded after their chemical energy is exhausted. They are also called dry cells because they do not contain any liquid electrolyte .
- Secondary cells are batteries that can be recharged by applying an external electric current that reverses the chemical reaction. They are also called wet cells because they usually contain a liquid electrolyte .
- Primary cells are further classified into two subcategories based on their application: household batteries and industrial batteries.
- Household batteries are the types of batteries that are more likely to be known to the common man. They find uses in a wide range of household appliances (such as torches, clocks, and cameras). Some examples of household batteries are zinc-carbon batteries, alkaline batteries, lithium batteries, and button cells .
- Industrial batteries are built to serve heavy-duty requirements. They are used in applications that require high power, long life, and reliability. Some examples of industrial batteries are lead-acid batteries, nickel-cadmium batteries, nickel-metal hydride batteries, and lithium-ion batteries .
- Secondary cells are further classified into two subcategories based on their application: vehicle batteries and portable batteries.
- Vehicle batteries are more user-friendly and a less complicated version of the industrial batteries. They are used to provide electrical power to automobiles, motorcycles, and other vehicles. Some examples of vehicle batteries are lead-acid batteries, nickel-cadmium batteries, nickel-metal hydride batteries, and lithium-ion batteries .
- Portable batteries are the types of batteries that are used in devices that can be carried around easily. They are used in applications that require low power, light weight, and long life. Some examples of portable batteries are nickel-cadmium batteries, nickel-metal hydride batteries, lithium-ion batteries, and lithium-polymer batteries .

Lead Acid Battery

- A lead acid battery is a type of **rechargeable battery** that uses **lead** and **lead oxide** as electrodes and **sulfuric acid** as electrolyte.

- It was the **first** type of rechargeable battery ever invented by **Gaston Planté** in **1859**.
- It has a relatively **low energy density** compared to modern batteries, but it is **cheap**, **reliable**, and **easy to manufacture** .
- It is widely used in **automotive**, **industrial**, and **backup power** applications .
- It operates on the principle of **reversible chemical reactions** between the electrodes and the electrolyte during **charging** and **discharging** .
- The main reactions are:
 - Charging: $\text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4 \rightarrow 2\text{PbSO}_4 + 2\text{H}_2\text{O}$
 - Discharging: $2\text{PbSO}_4 + 2\text{H}_2\text{O} \rightarrow \text{Pb} + \text{PbO}_2 + 2\text{H}_2\text{SO}_4$
- The **voltage** of a single lead acid cell is about **2 V**, and the **capacity** depends on the **size** and **design** of the battery.
- The **efficiency** of a lead acid battery is about **75-85%**, and the **lifetime** is about **3-5 years** or **500-1000 cycles** .
- The main **advantages** of lead acid batteries are:
 - Low cost
 - High power output
 - Good low-temperature performance
 - Simple and safe charging
 - Recyclable
- The main **disadvantages** of lead acid batteries are:
 - Low energy density
 - Heavy and bulky
 - Sulfation and corrosion
 - Water loss and maintenance
 - Environmental hazards

Corrosion: Introduction to corrosion, Types of corrosion, Cause of corrosion, Corrosion

- Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment.
- Corrosion can cause loss of material, mechanical strength, functionality, and aesthetic appeal of the affected material.
- Corrosion can also pose safety, health, and environmental hazards if not prevented or controlled.
- Corrosion can be classified into different types based on the nature and mechanism of the corrosion process, such as:

- Uniform or general corrosion: The metal surface is uniformly attacked by the corrosive medium, resulting in a uniform loss of material.
- Galvanic or bimetallic corrosion: Two dissimilar metals are in contact with each other and an electrolyte, creating a galvanic cell that causes one metal to corrode preferentially (the anode) while the other metal is protected (the cathode).
- Pitting corrosion: The metal surface is locally attacked by the corrosive medium, resulting in small pits or holes that can penetrate deep into the metal.
- Crevice corrosion: The metal surface is attacked by the corrosive medium in narrow spaces or crevices, where the oxygen concentration is low and the acidity is high.
- Intergranular corrosion: The metal surface is attacked by the corrosive medium along the grain boundaries of the metal, where the microstructure or composition may differ from the bulk metal.
- Stress corrosion cracking: The metal surface is attacked by the corrosive medium in the presence of tensile stress, resulting in cracks that can propagate and cause failure of the metal.
- Corrosion fatigue: The metal surface is attacked by the corrosive medium in the presence of cyclic stress, resulting in cracks that can initiate and grow due to the combined effect of corrosion and fatigue.
- Erosion corrosion: The metal surface is attacked by the corrosive medium in the presence of fluid flow, resulting in material loss due to the mechanical abrasion and impact of the fluid particles.
- Microbiologically influenced corrosion: The metal surface is attacked by the corrosive medium in the presence of microorganisms, such as bacteria, fungi, or algae, that can influence the corrosion process by producing corrosive substances, forming biofilms, or altering the local environment.
- The cause of corrosion can be attributed to various factors, such as:
 - The nature and composition of the metal and its environment, which determine the thermodynamic tendency and the kinetic rate of the corrosion reaction.
 - The physical and chemical properties of the metal and its environment, such as temperature, pressure, pH, salinity, oxygen concentration, etc., which affect the corrosion reaction and its products.
 - The design and fabrication of the metal and its environment, such as shape, size, surface finish, welding, coating, etc., which influence the corrosion behavior and resistance of the metal.
 - The operational and environmental conditions of the metal and its environment, such as stress, strain, vibration, wear, erosion, etc., which induce or enhance the corrosion process and its damage.
- Corrosion can be prevented or controlled by various methods, such as:
 - Material selection: Choosing a metal that is inherently resistant to corrosion or compatible with the corrosive environment, or using a

non-metallic material that is immune to corrosion.

- **Protective coating:** Applying a layer of material, such as paint, plastic, metal, etc., that can isolate the metal from the corrosive environment, or provide sacrificial protection by corroding instead of the metal.
- **Cathodic protection:** Applying an external electric current or a sacrificial anode to the metal, which can reduce or eliminate the corrosion of the metal by making it the cathode of the galvanic cell.
- **Corrosion inhibitors:** Adding a substance, such as an organic compound, a metal ion, or an oxide, to the corrosive environment, which can interfere with the corrosion reaction by forming a protective film, reducing the aggressiveness of the environment, or inhibiting the cathodic or anodic reaction.
- **Environmental modification:** Changing the physical or chemical parameters of the corrosive environment, such as temperature, pressure, pH, salinity, oxygen concentration, etc., which can reduce or eliminate the corrosion of the metal by shifting the thermodynamic equilibrium or the kinetic rate of the corrosion reaction.
- **Design improvement:** Modifying the shape, size, surface finish, welding, coating, etc., of the metal and its environment, which can improve the corrosion behavior and resistance of the metal by avoiding or minimizing the factors that induce or enhance the corrosion process and its damage.

Prevention and Control of Corrosion

Corrosion is the deterioration of a material, usually a metal, due to a chemical or electrochemical reaction with its environment. Corrosion can cause loss of material, loss of function, loss of aesthetics, and loss of safety. Therefore, corrosion prevention and control are important aspects of engineering and maintenance.

Some of the methods of corrosion prevention and control are:

- **Material selection:** Choosing a material that is resistant to corrosion in a given environment, or that can form a protective layer or coating on its surface. For example, stainless steel, aluminum, copper, and titanium are some of the metals that have good corrosion resistance in various environments.
- **Coating:** Applying a layer of another material on the surface of the metal to protect it from corrosion. The coating can be metallic, such as zinc, nickel, or chromium, or non-metallic, such as paint, plastic, or rubber. The coating should be compatible with the metal and the environment, and should be able to adhere well and resist wear and damage.
- **Cathodic protection:** Applying an external electric current to the metal to make it the cathode of an electrochemical cell, and thus prevent it from corroding. The current can be supplied by a sacrificial anode, such as mag-

nesium or zinc, which corrodes instead of the metal, or by an impressed current from a power source. Cathodic protection is commonly used for pipelines, tanks, ships, and offshore structures.

- **Anodic protection:** Applying an external electric current to the metal to make it the anode of an electrochemical cell, and thus form a protective oxide layer on its surface. The current can be supplied by a power source or by a galvanic couple with another metal. Anodic protection is used for metals that can passivate, such as stainless steel, in environments that are oxidizing, such as nitric acid or sulfuric acid.
- **Corrosion inhibitors:** Adding chemicals to the environment that can reduce the rate of corrosion by interfering with the corrosion reaction. The inhibitors can act by adsorbing on the metal surface and forming a barrier, by scavenging the corrosive agents, such as oxygen or chloride, or by modifying the pH or the conductivity of the environment. Corrosion inhibitors are used for various applications, such as cooling water systems, oil and gas production, and metal processing.
- **Environmental control:** Modifying the environment to make it less corrosive to the metal. This can include reducing the temperature, humidity, salinity, acidity, or oxygen content of the environment, or removing the contaminants, such as dust, dirt, or microbes, that can accelerate corrosion. Environmental control can also involve isolating the metal from the environment, such as by using seals, gaskets, or barriers.

Issues in specific industries (Power generation, Chemical)

- Power generation is the process of converting primary energy sources, such as fossil fuels, nuclear energy, renewable energy, etc., into electricity.
- Chemical industry is the branch of manufacturing that produces chemicals and chemical products, such as plastics, fertilizers, pharmaceuticals, etc., from raw materials, such as oil, gas, coal, minerals, etc.

Issues in power generation

- Some of the major issues in power generation are:
 - Environmental impact: Power generation from fossil fuels and nuclear energy produces greenhouse gas emissions, air pollutants, radioactive waste, etc., that contribute to global warming, climate change, acid rain, smog, etc., and pose health and safety risks to humans and ecosystems.
 - Resource depletion: Power generation from fossil fuels and nuclear energy relies on finite and non-renewable resources, such as coal, oil, gas, uranium, etc., that are subject to scarcity, price volatility, geopolitical conflicts, etc., and may run out in the future.

- Energy security: Power generation from fossil fuels and nuclear energy depends on the availability and accessibility of the resources, which may be affected by natural disasters, wars, sanctions, trade disputes, etc., and may cause disruptions, shortages, or blackouts in the electricity supply.
- Cost and efficiency: Power generation from fossil fuels and nuclear energy involves high capital, operating, and maintenance costs, as well as low efficiency and high losses in the conversion, transmission, and distribution of electricity.

Issues in chemical industry

- Some of the major issues in chemical industry are:
 - Environmental impact: Chemical industry produces and uses hazardous substances, such as toxic, flammable, explosive, corrosive, etc., that may cause pollution, contamination, accidents, etc., in the air, water, soil, etc., and harm humans and ecosystems.
 - Health and safety: Chemical industry exposes workers and consumers to various risks, such as exposure, inhalation, ingestion, contact, etc., to harmful chemicals, that may cause acute or chronic diseases, injuries, disabilities, or death.
 - Regulation and compliance: Chemical industry faces strict and complex regulations and standards, such as REACH, TSCA, OSHA, etc., from various authorities and agencies, such as EPA, FDA, ECHA, etc., that aim to ensure the quality, safety, and sustainability of the chemical products and processes, but also impose challenges, costs, and liabilities to the industry.
 - Innovation and competition: Chemical industry requires constant research and development, innovation, and adaptation, to meet the changing demands, preferences, and expectations of the customers, markets, and society, as well as to cope with the emerging technologies, trends, and opportunities, such as biotechnology, nanotechnology, green chemistry, etc., but also to face the increasing competition, pressure, and threats from the domestic and global competitors.

Processing industry, Oil & gas industry and Pulp & paper industries

Processing industry

- A processing industry is a type of industry that transforms raw materials or intermediate products into finished goods or services.
- Examples of processing industries include food and beverage, textile, chemical, pharmaceutical, metal, and petroleum industries.

- Processing industries can be classified into two main types: continuous and batch processing.
 - Continuous processing involves a constant flow of materials through a series of operations, such as in oil refining or paper making.
 - Batch processing involves processing discrete quantities of materials in separate batches, such as in baking or brewing.

Oil & gas industry

- The oil & gas industry is a global sector that explores, produces, transports, refines, and markets crude oil, natural gas, and petroleum products.
- The oil & gas industry can be divided into three main segments: upstream, midstream, and downstream.
 - Upstream involves the exploration and production of oil and gas from reservoirs, wells, and fields.
 - Midstream involves the transportation and storage of oil and gas, such as by pipelines, tankers, and terminals.
 - Downstream involves the refining and processing of oil and gas into fuels, lubricants, chemicals, and other products, as well as the distribution and marketing of these products to end-users.
- The oil & gas industry is influenced by various factors, such as supply and demand, geopolitics, environmental regulations, technological innovations, and price fluctuations.

Pulp & paper industry

- The pulp & paper industry is a sector that produces pulp, paper, and paperboard from wood and other raw materials, such as recycled paper, agricultural residues, and non-wood fibers.
- The pulp & paper industry can be divided into four main segments: pulp, paper, paperboard, and tissue.
 - Pulp is the fibrous material that is produced by mechanically or chemically separating wood or other fibers, and is used as the main ingredient for paper and paperboard.
 - Paper is a thin sheet of material that is made by forming a pulp slurry into a web and drying it, and is used for various purposes, such as writing, printing, packaging, and hygiene.
 - Paperboard is a thicker and stiffer type of paper that is made by laminating or compressing multiple layers of paper, and is used for applications that require more strength and durability, such as boxes, cartons, and containers.
 - Tissue is a soft and absorbent type of paper that is made by creping a thin sheet of paper, and is used for products that require high softness and absorbency, such as toilet paper, facial tissue, and napkins.
- The pulp & paper industry is affected by various factors, such as consumer demand, environmental sustainability, energy efficiency, and digi-

talization.

Chemistry of Engineering Materials:

- Engineering materials are materials that are used as raw materials for any sort of construction or manufacturing in an organized way of engineering application .
- Engineering materials can be classified into four main categories: metals, ceramics, polymers, and composites.
- The chemical properties of engineering materials refer to the elements that are combined together to form the material and how they affect its behavior and performance.
- Some of the important chemical properties of engineering materials are:
 - Chemical composition: The chemical composition of a material determines its strength, hardness, ductility, brittleness, corrosion resistance, weldability, etc. For example, steel is an alloy of iron and carbon, and its properties depend on the amount and distribution of carbon in the material.
 - Acidity or alkalinity: The acidity or alkalinity of a material is measured by its pH value, which ranges from 0 to 14. A material with a pH value of 7 is neutral, while a material with a pH value below 7 is acidic and a material with a pH value above 7 is alkaline. The acidity or alkalinity of a material affects its corrosion resistance, electrical conductivity, and biological compatibility. For example, stainless steel is more resistant to corrosion than carbon steel because it has a higher pH value.
 - Reactivity: The reactivity of a material is its tendency to undergo chemical reactions with other substances, such as oxygen, water, acids, bases, etc. The reactivity of a material affects its stability, durability, and safety. For example, magnesium is a highly reactive metal that can ignite easily when exposed to air or water, while gold is a very inert metal that does not react with most substances.
 - Energy conversion: Some engineering materials can convert one form of energy into another, such as electrical, thermal, mechanical, chemical, or nuclear energy. These materials are called active materials and they are used in various applications, such as solar cells, batteries, catalysts, and superconducting magnets. The energy conversion of a material depends on its chemical structure, composition, and interactions with other materials. For example, a solar cell is a device that converts light energy into electrical energy by using a semiconductor material that absorbs photons and generates electrons.

Cement; Constituents, manufacturing, hardening and setting, deterioration of cement, Plaster

- Cement is a binding material that sets, hardens, and adheres to other materials to bind them together.
- The main constituents of cement are calcium, silicon, aluminum, iron and other ingredients that are combined in different proportions depending on the type and grade of cement.
- The manufacturing of cement involves four main stages: raw material preparation, clinker production, clinker grinding, and cement packing.
 - Raw material preparation involves crushing, grinding, and blending of limestone, clay, shale, iron ore, and other materials to obtain a homogeneous mixture of the required composition.
 - Clinker production involves heating the raw material mixture in a rotary kiln at a high temperature (around 1400-1500°C) to form clinker, which is a nodular material consisting of calcium silicates and aluminates.
 - Clinker grinding involves adding gypsum and other additives to the clinker and grinding it to a fine powder, which is the final product known as cement.
 - Cement packing involves filling the cement powder into bags or silos for storage or transportation.
- The hardening and setting of cement is due to the formation of interlocking crystals reinforced by rigid gels formed by the hydration and hydrolysis of the constitutional compounds.
 - The hydration of cement is a chemical reaction between water and the cement compounds that releases heat and forms new compounds.
 - The hydrolysis of cement is a chemical reaction between water and the cement compounds that consumes water and forms new compounds.
 - The initial and final setting times of cement are of practical importance, as they determine the length of time in which cement mixes (concrete) remain plastic and workable, and the length of time after which they can bear load and resist deformation.
 - The setting and hardening of cement are influenced by several factors, such as the water-cement ratio, the temperature, the curing conditions, the type and amount of additives, and the fineness of cement.
- The deterioration of cement is the loss of strength, durability, or appearance of cement or concrete due to various causes, such as chemical attack, corrosion of embedded metals, physical damage, or environmental factors.
 - Chemical attack is the reaction of cement or concrete with some natural or artificial chemical agents, such as acids, sulfates, chlorides, carbon dioxide, etc., that can alter the composition, structure, or properties of the cementitious materials .
 - Corrosion of embedded metals is the oxidation of steel reinforcement

- or other metal components in concrete due to the presence of moisture, oxygen, and chloride ions, which can cause cracking, spalling, or delamination of concrete.
- Physical damage is the result of mechanical stress, abrasion, erosion, impact, or thermal shock on cement or concrete, which can cause cracking, chipping, or scaling of the surface or the interior of the material.
- Environmental factors are the exposure of cement or concrete to extreme temperatures, humidity, freeze-thaw cycles, ultraviolet radiation, or biological agents, which can affect the performance, durability, or appearance of the material.
- Plaster is a thin layer of cement or other material applied to a wall, ceiling, or other surface to provide a smooth, decorative, or protective finish.
 - Plaster can be made of different materials, such as lime, gypsum, cement, clay, or synthetic polymers, depending on the desired properties, such as strength, hardness, water resistance, fire resistance, or aesthetics.
 - Plaster can be applied by hand or by machine, using a trowel, a spray gun, or a roller, depending on the type and thickness of the plaster and the surface to be covered.
 - Plaster can be finished in different ways, such as by sanding, polishing, painting, or texturing, to achieve a smooth, glossy, matte, or patterned surface.

Paris

Paris is the capital and most populous city of France, with an estimated population of 2,165,423 residents in 2019 in an area of more than 105 km² (41 sq mi), making it the fourth-most populated city in the European Union as well as the 30th most densely populated city in the world in 2022. Paris is a global center for art, fashion, gastronomy, and culture, and is often regarded as the most romantic city in the world.

Some of the main points to know about Paris are:

- Paris is situated in the north-central part of France, along the Seine River, which divides the city into two parts: the Right Bank and the Left Bank.
- Paris has a rich and turbulent history, dating back to the 3rd century BCE, when it was founded by a Celtic tribe called the Parisii. It became the capital of the Frankish kingdom in the 6th century CE, and later the seat of the French monarchy, the French Revolution, and the French Empire.
- Paris is home to many iconic landmarks, such as the Eiffel Tower, the Arc de Triomphe, the Notre-Dame Cathedral, the Louvre Museum, and the Palace of Versailles. It also has many museums, art galleries, parks, gardens, and monuments that showcase its cultural and artistic heritage.
- Paris is known for its cuisine, which includes dishes such as croissants,

macarons, frog legs, steak and frites, eclairs, and French onion soup. It also has a reputation for its wine, cheese, and chocolate, which are popular souvenirs for tourists.

- Paris is a major hub for fashion, hosting the Paris Fashion Week twice a year, and being the birthplace of many famous designers, such as Coco Chanel, Christian Dior, and Yves Saint Laurent. It also has many boutiques, department stores, and flea markets that offer a variety of styles and trends.
- Paris is a vibrant and diverse city, with many neighborhoods, districts, and quarters that have their own character and atmosphere. Some of the most famous ones are Montmartre, the Latin Quarter, the Marais, and the Champs-Élysées.
- Paris is a popular destination for travelers, attracting more than 30 million visitors annually. It offers many attractions, activities, and events for different interests and preferences, such as sightseeing, shopping, dining, nightlife, entertainment, and romance.

Unit-4: 8

- This unit covers the topic of **linear equations** and their applications in various contexts.
- A linear equation is an equation of the form $\mathbf{ax} + \mathbf{b} = \mathbf{c}$, where $\mathbf{a}$, $\mathbf{b}$, and $\mathbf{c}$ are constants and $\mathbf{x}$ is a variable.
- The solution of a linear equation is the value of $\mathbf{x}$ that makes the equation true. For example, the solution of $\mathbf{2x} + \mathbf{3} = \mathbf{11}$ is $\mathbf{x} = \mathbf{4}$.
- To solve a linear equation, we can use the following steps:
 - Simplify both sides of the equation by combining like terms and applying the distributive property if needed.
 - Isolate the variable on one side of the equation by adding, subtracting, multiplying, or dividing both sides by the same number.
 - Check the solution by substituting it back into the original equation and verifying that both sides are equal.
- A linear equation can represent a real-world situation, such as a relationship between two quantities, a cost function, a rate of change, or a balance point. For example, the equation $\mathbf{y} = \mathbf{3x} + \mathbf{5}$ can represent the cost $\mathbf{y}$ of renting a car for $\mathbf{x}$ days at \$3 per day plus a \$5 fee.
- To model a real-world situation with a linear equation, we can use the following steps:
 - Identify the two quantities that are related and assign them variables, such as $\mathbf{x}$ and $\mathbf{y}$.
 - Find the slope of the line that represents the rate of change between the two quantities, such as $\mathbf{m}$.
 - Find the y-intercept of the line that represents the initial or fixed value of one quantity, such as $\mathbf{b}$.
 - Write the equation of the line in the form $\mathbf{y} = \mathbf{mx} + \mathbf{b}$ or $\mathbf{ax} + \mathbf{b} = \mathbf{c}$.

- Interpret the meaning of the slope and the y-intercept in the context of the problem.
- A linear equation can also be used to find the value of one quantity given the value of another quantity, or to compare two different situations. For example, we can use the equation $y = 3x + 5$ to find the cost of renting a car for 7 days, or to compare the cost of renting a car with another company that charges \$4 per day plus a \$2 fee.

Water Technology: Sources and impurities of water, Hardness of water, Boiler troubles

Sources and impurities of water

- Water is a vital resource for life and various human activities. It is found abundantly in the universe, but not all water sources are pure and suitable for consumption or industrial use.
- Water sources can be classified into three main categories: sky sources, surface sources, and subsoil sources.
 - Sky sources include rainfall, snowfall, hail, and dew. These sources of water are initially pure, but they can absorb impurities from the air, such as dust, gases, bacteria, and pollutants.
 - Surface sources include rivers, lakes, streams, canals, and oceans. These sources of water are contaminated by the impurities present on the earth's surface, such as soil, minerals, organic matter, microorganisms, and human activities.
 - Subsoil sources include wells, springs, and aquifers. These sources of water are filtered by the soil and rocks, but they can also contain dissolved salts, metals, and chemicals that leach from the underground layers.
- Water impurities can be classified into two main types: dissolved and suspended solids.
 - Dissolved solids are substances that are soluble in water, such as salts, acids, bases, gases, and minerals. They can affect the taste, odor, color, and hardness of water.
 - Suspended solids are substances that are insoluble or partially soluble in water, such as clay, sand, silt, organic matter, and microorganisms. They can affect the turbidity, clarity, and sedimentation of water.
- Water impurities can have various effects on human health and the environment, depending on their nature, concentration, and exposure. Some common water contaminants and their potential risks are:
 - Bacteria, viruses, and parasites: These microorganisms can cause infections, diarrhea, dysentery, cholera, typhoid, and other diseases.
 - Nitrates and nitrites: These compounds can result from the decomposition of organic matter, fertilizers, and animal waste. They can cause methemoglobinemia, or blue baby syndrome, in infants and pregnant women.

- Lead, arsenic, mercury, and cadmium: These metals can leach from pipes, mining, and industrial activities. They can cause neurological, cardiovascular, renal, and reproductive problems, as well as cancer.
- Chlorine, chloramines, and disinfection byproducts: These chemicals are used to disinfect water, but they can also react with organic matter and form harmful compounds, such as trihalomethanes and haloacetic acids. They can cause respiratory, skin, and eye irritation, as well as cancer.
- Fluoride: This mineral is naturally present in some water sources, but it can also be added to prevent tooth decay. However, excessive fluoride intake can cause dental and skeletal fluorosis, which can damage the teeth and bones.
- Pesticides, herbicides, and insecticides: These substances are used in agriculture and gardening, but they can also contaminate water sources through runoff, leaching, and drift. They can affect the nervous, endocrine, and immune systems, as well as cause cancer.
- Pharmaceuticals and personal care products: These substances are used for medical and cosmetic purposes, but they can also enter water sources through human and animal excretion, disposal, and wastewater treatment. They can interfere with the hormonal, reproductive, and developmental functions of humans and wildlife.

Techniques for Water Softening

Water softening is the process of removing calcium, magnesium, and other metal ions from hard water, which can cause problems such as scaling, soap scum, corrosion, and reduced efficiency of appliances and heating systems. There are different techniques for water softening, each with its own advantages and disadvantages. Some of the most common techniques are:

- **Lime-Soda Process:** This is a chemical method that involves adding lime (calcium hydroxide) and soda ash (sodium carbonate) to hard water to precipitate the calcium and magnesium ions as insoluble carbonates. The resulting soft water has a high pH and may contain some residual hardness. This method is suitable for large-scale industrial applications, but not for domestic use.
- **Zeolite Process:** This is a physical method that uses natural or synthetic zeolites, which are porous minerals that can exchange sodium ions for calcium and magnesium ions in hard water. The zeolites are regenerated by washing them with a brine solution. The resulting soft water has a low pH and may contain some sodium. This method is suitable for domestic and commercial use, but not for very hard water or water with high iron content.
- **Ion Exchange Process:** This is a physical method that uses resin beads, which are synthetic polymers that can exchange sodium or potassium ions

for calcium and magnesium ions in hard water. The resin beads are regenerated by washing them with a salt solution. The resulting soft water has a low pH and may contain some sodium or potassium. This method is suitable for domestic and commercial use, and can handle very hard water and water with high iron content.

- **Reverse Osmosis Process:** This is a physical method that uses a semi-permeable membrane to separate the dissolved salts and minerals from hard water. The water is forced through the membrane under high pressure, leaving behind the impurities. The resulting soft water has a neutral pH and may contain some dissolved gases. This method is suitable for domestic and commercial use, and can produce very pure water. However, it requires a lot of energy and produces a lot of wastewater.

These are some of the techniques for water softening. Each technique has its own benefits and drawbacks, and the choice of technique depends on factors such as the source and quality of water, the desired level of softness, the cost and availability of materials and equipment, and the environmental impact.

Process, Determination of Hardness and Alkalinity, Numerical Problems

- **Process** is a series of actions or steps taken to achieve a desired result or outcome. In the context of water treatment, process refers to the methods and techniques used to purify water for various purposes, such as drinking, irrigation, industrial use, etc.
- **Hardness** is a measure of the concentration of dissolved calcium and magnesium ions in water. Hard water can cause scaling, corrosion, and reduced efficiency of boilers, pipes, and appliances. Hardness can be expressed in various units, such as milligrams per liter (mg/L) of calcium carbonate (CaCO_3), grains per gallon (gpg), or degrees of hardness (dH).
- **Alkalinity** is a measure of the capacity of water to neutralize acids. Alkalinity is mainly due to the presence of bicarbonates, carbonates, and hydroxides in water. Alkalinity can also be expressed in various units, such as mg/L of CaCO_3 , meq/L, or dH.
- **Determination of Hardness and Alkalinity** can be done by various methods, such as titration, colorimetry, or ion exchange. One of the most common methods is the EDTA titration method, which involves adding a standard solution of ethylenediaminetetraacetic acid (EDTA) to a water sample until a color change occurs, indicating the end point. The amount of EDTA used is proportional to the hardness or alkalinity of the water sample. The indicator used for hardness titration is Eriochrome Black T (EBT), which changes from red to blue when all the calcium and magnesium ions are complexed by EDTA. The indicator used for alkalinity

titration is phenolphthalein, which changes from colorless to pink when the pH reaches 8.3, indicating the end point for bicarbonate alkalinity, and methyl orange, which changes from yellow to orange when the pH reaches 4.5, indicating the end point for total alkalinity.

- **Numerical Problems** are problems that require the application of mathematical formulas and calculations to solve. Some examples of numerical problems related to hardness and alkalinity are:
 - A water sample has a total hardness of 250 mg/L as CaCO_3 and a calcium hardness of 150 mg/L as CaCO_3 . What is the magnesium hardness of the water sample?
 - * Solution: Magnesium hardness = Total hardness - Calcium hardness
 - * Magnesium hardness = $250 - 150 = 100$ mg/L as CaCO_3
 - A water sample has a total alkalinity of 300 mg/L as CaCO_3 and a bicarbonate alkalinity of 200 mg/L as CaCO_3 . What is the carbonate alkalinity and the hydroxide alkalinity of the water sample?
 - * Solution: Carbonate alkalinity = Total alkalinity - Bicarbonate alkalinity
 - * Carbonate alkalinity = $300 - 200 = 100$ mg/L as CaCO_3
 - * Hydroxide alkalinity = Carbonate alkalinity - $2 \times$ Bicarbonate alkalinity
 - * Hydroxide alkalinity = $100 - 2 \times 200 = -300$ mg/L as CaCO_3
 - * Since hydroxide alkalinity cannot be negative, it is assumed to be zero, and the excess carbonate alkalinity is converted to bicarbonate alkalinity.
 - * Bicarbonate alkalinity = $200 + 300 = 500$ mg/L as CaCO_3
 - * Carbonate alkalinity = 0 mg/L as CaCO_3
 - * Hydroxide alkalinity = 0 mg/L as CaCO_3
 - A water sample requires 20 mL of 0.01 M EDTA to reach the end point with EBT indicator for hardness titration. What is the total hardness of the water sample in mg/L as CaCO_3 and in dH?
 - * Solution: Total hardness = Volume of EDTA $\times$ Molarity of EDTA $\times$ Equivalent weight of $\text{CaCO}_3 \times 1000$ / Volume of water sample
 - * Equivalent weight of CaCO_3 = Molecular weight of CaCO_3 / 2 = $100 / 2 = 50$
 - * Assuming the volume of water sample is 100 mL, the total hardness is:
 - * Total hardness = $20 \times 0.01 \times 50 \times 1000 / 100 = 200$ mg/L as CaCO_3
 - * To convert mg/L as CaCO_3 to dH, divide by 17.8
 - * Total hardness = $200 / 17.8 = 11.24$ dH

Fuels and Combustion: Definition, Classification, Characteristics of a good fuel, Calorific

- **Definition:** Fuel is a material that produces heat and energy on combustion. Combustion is a chemical reaction in which a fuel combines with oxygen and releases heat and light.
- **Classification:** Fuels can be classified into three main types based on their physical state: solid, liquid and gaseous. Examples of solid fuels are coal, wood and charcoal. Examples of liquid fuels are petrol, diesel and kerosene. Examples of gaseous fuels are natural gas, biogas and hydrogen.
- **Characteristics of a good fuel:** A good fuel should have the following properties :
 - It should be cheap, easily available and readily combustible.
 - It should have a high calorific value, which is the quantity of heat produced by the combustion of a unit mass of fuel.
 - It should not produce harmful gases or residues that pollute the environment, such as carbon monoxide, sulphur dioxide and ash.
 - It should be dry and should have less moisture content, as dry fuel increases its calorific value and reduces the cost of transportation and storage.
 - It should have a low ignition temperature, which is the minimum temperature at which a fuel catches fire spontaneously.
 - It should have a moderate rate of combustion, which is the speed at which a fuel burns. A fuel that burns too fast or too slow may cause problems in controlling the flame and heat output.
 - It should have a high flame temperature, which is the maximum temperature attained by the flame during combustion. A high flame temperature indicates a high efficiency of fuel utilization.
- **Calorific:** Calorific is an adjective that means relating to or producing heat or calories. Calorific value is a measure of the heat energy released by a fuel on combustion. It is usually expressed in kilojoules per kilogram (kJ/kg) or megajoules per kilogram (MJ/kg) for solid and liquid fuels, and kilojoules per cubic meter (kJ/m³) or megajoules per cubic meter (MJ/m³) for gaseous fuels. The higher the calorific value of a fuel, the more heat it can produce per unit mass or volume. Calorific value can be determined by using a device called a bomb calorimeter, which measures the heat released by burning a known mass of fuel in a closed chamber under constant pressure and temperature. Calorific value can also be calculated by using the chemical formula of the fuel and the heat of formation of the products of combustion. Calorific value can vary depending on the moisture content, purity and composition of the fuel.

Values, Gross & Net calorific value, Determination of calorific value by Bomb Calorimeter

- Values are the numerical quantities that represent the worth or importance of something.
- Calorific value is the amount of heat or energy produced by a fuel during complete combustion. It is measured in units of energy per unit mass or volume of fuel, such as calories per gram or joules per kilogram.
- Gross calorific value (GCV) is the total amount of heat released when a fuel is burned completely, including the latent heat of vaporization of water vapors produced by the combustion. It is also known as the higher heating value (HHV) of the fuel.
- Net calorific value (NCV) is the amount of heat released when a fuel is burned completely, excluding the latent heat of vaporization of water vapors produced by the combustion. It is also known as the lower heating value (LHV) of the fuel.
- The formula for calculating the net calorific value from the gross calorific value is:

Net calorific value = gross calorific value - latent heat of water vapors

where the latent heat of water vapors is the product of the mass of water vapors and the specific latent heat of vaporization of water at a given temperature.

- A bomb calorimeter is an instrument used to determine the calorific value of a fuel by measuring the heat transferred to a known mass of water in a closed and insulated container. The basic principle of a bomb calorimeter is:

Heat released by fuel = heat gained by water + heat gained by calorimeter

where the heat released by fuel is the product of the mass of fuel and the calorific value of fuel, and the heat gained by water and calorimeter are the products of their respective masses, specific heat capacities, and temperature changes.

- The construction and working of a bomb calorimeter are as follows:
 - A bomb calorimeter consists of a strong steel vessel called the bomb, which can withstand high pressure and temperature. The bomb has a lid with two electrodes and a fuse wire attached to it. The bomb is filled with pure oxygen gas at a high pressure.
 - A known mass of fuel sample is placed in a capsule and connected to the fuse wire. The capsule is then placed in the bomb and the lid is tightly screwed. The bomb is then immersed in a known mass of

water in a calorimeter vessel, which is insulated from the surroundings by an air jacket or a water jacket.

- The temperature of the water and the calorimeter vessel is measured before and after the combustion. The combustion is initiated by passing an electric current through the fuse wire, which ignites the fuel sample. The heat released by the combustion is transferred to the water and the calorimeter vessel, raising their temperatures.
- The calorific value of the fuel sample can be calculated by using the formula:

Calorific value of fuel = (heat gained by water + heat gained by calorimeter) / mass of fuel

where the heat gained by water and calorimeter are the products of their respective masses, specific heat capacities, and temperature changes. The specific heat capacity of the calorimeter can be determined by a calibration experiment using a standard fuel of known calorific value.

Theoretical calculation of calorific value by Dulong's method, Ranking of Coal, Analysis of coal

- Calorific value of coal is the amount of heat released by the complete combustion of a unit mass of coal. It is an important parameter for evaluating the quality and efficiency of coal.
- Dulong's method is a formula that estimates the calorific value of coal from its ultimate analysis, which is the percentage of carbon, hydrogen, oxygen, nitrogen and sulfur in coal.
- Dulong's formula is given by:

$$CV = 337C + 1442(H - O/8) + 93S$$

where CV is the calorific value in kJ/kg, and C, H, O, S are the percentages of carbon, hydrogen, oxygen and sulfur in coal, respectively.

- Dulong's formula is based on the assumption that the heat of combustion of an organic compound is nearly equal to the heats of combustion of the elements in it, multiplied by their percentage content in the compound .
- Dulong's formula has some limitations, such as:
 - It does not account for the moisture and ash content of coal, which affect the calorific value.
 - It does not account for the variation of the heats of combustion of the elements with temperature and pressure.

- It does not account for the formation of complex compounds during coal combustion, such as CO, CO₂, H₂O, SO₂, etc.
 - It does not account for the variation of the calorific value of coal with its rank and type.
- Therefore, Dulong's formula is only an approximation and should be used with caution. Some modifications and corrections have been proposed to improve the accuracy of Dulong's formula, such as the modified Dulong formula and the Vandralek formula .
- Ranking of coal is the classification of coal according to its degree of metamorphism, or the change in its physical and chemical properties due to the action of heat and pressure over time. The rank of coal reflects its maturity, quality and potential uses.
- The main ranks of coal are, from lowest to highest:
 - Peat: The lowest rank of coal, formed from partially decomposed plant matter in wetlands. It has a high moisture content, low calorific value, and high volatile matter. It is not a true coal, but a precursor of coal.
 - Lignite: The lowest rank of true coal, formed from peat under moderate pressure and temperature. It has a brown color, a high moisture content, a low calorific value, and a high volatile matter. It is also called brown coal.
 - Sub-bituminous: A rank of coal intermediate between lignite and bituminous, formed from lignite under higher pressure and temperature. It has a black color, a lower moisture content, a higher calorific value, and a lower volatile matter than lignite. It is also called black lignite.
 - Bituminous: A high rank of coal, formed from sub-bituminous coal under higher pressure and temperature. It has a black color, a low moisture content, a high calorific value, and a moderate volatile matter. It is the most abundant and widely used rank of coal. It is also called soft coal.
 - Anthracite: The highest rank of coal, formed from bituminous coal under very high pressure and temperature. It has a shiny black color, a very low moisture content, a very high calorific value, and a very low volatile matter. It is the hardest and cleanest rank of coal. It is also called hard coal or stone coal.
- The rank of coal can be determined by various parameters, such as:
 - Fixed carbon: The percentage of carbon in coal that remains after the volatile matter is driven off by heating. It increases with the rank of coal.
 - Volatile matter: The percentage of coal that is converted into gases and vapors by heating. It decreases with the rank of coal.

- Moisture: The percentage of water in coal. It decreases with the rank of coal.
- Ash: The percentage of inorganic matter in coal that remains after combustion. It varies with the rank of coal and the source of coal.
- Calorific value: The amount of heat released by the combustion of a unit mass of coal. It increases with the rank of coal.
- Carbon-hydrogen ratio: The ratio of the percentage of carbon to the percentage of hydrogen in coal. It increases with the rank of coal.
- Analysis of coal is the process of

Proximate and Ultimate Analysis Method

- Proximate analysis is the technique used to analyze the compounds in a mixture, such as coal, biomass, or biogas.
- Ultimate analysis is the technique used to analyze the elements present in a compound, such as carbon, hydrogen, nitrogen, sulfur, and oxygen.
- Proximate analysis consists of determination of ash content, moisture content, volatile matter, and fixed carbon on an as-received basis.
- Ultimate analysis provides a convenient method for reporting the major organic elemental composition of coal or other fuels.
- Proximate and ultimate analysis can be used to estimate the heating value, combustion characteristics, and environmental impacts of solid fuels and biogas .

Numerical Problems

- Example 1: A coal sample has the following proximate analysis (wt%): moisture 10, volatile matter 35, fixed carbon 45, ash 10. Calculate the mass of air required for complete combustion of 1 kg of coal, assuming that the coal contains only carbon and the air is dry. Assume that the excess air is 20%.
- Solution 1: The mass of carbon in 1 kg of coal is 0.45 kg. The stoichiometric air required for complete combustion of carbon is 2.67 kg of air per kg of carbon. The actual air required is 1.2 times the stoichiometric air, which is 3.2 kg of air per kg of carbon. Therefore, the mass of air required for complete combustion of 1 kg of coal is $3.2 \times 0.45 = 1.44$ kg.
- Example 2: A biogas sample has the following ultimate analysis (wt%): carbon 50, hydrogen 10, nitrogen 5, oxygen 35. Calculate the higher heating value (HHV) of the biogas in kJ/kg, assuming that the biogas contains only methane and carbon dioxide. Use the following values for the HHV of methane and carbon dioxide: 55.5 MJ/kg and -393.5 kJ/mol, respectively.
- Solution 2: The mass of methane in 1 kg of biogas is 0.5 kg. The mass of carbon dioxide in 1 kg of biogas is 0.5 kg. The HHV of biogas is the sum

of the HHV of methane and carbon dioxide, multiplied by their respective masses. Therefore, the HHV of biogas is $(55.5 \times 0.5) + (-393.5 \times 0.5 \times 44/12) = 24.8 \text{ MJ/kg}$.

Chemistry of Biogas

- Biogas is a mixture of gases produced by the anaerobic digestion of organic matter, such as animal manure, sewage, or crop residues.
- The main components of biogas are methane (CH_4) and carbon dioxide (CO_2), which account for about 50-70% and 30-50% of the biogas volume, respectively.
- Biogas may also contain small amounts of other gases, such as hydrogen sulfide (H_2S), ammonia (NH_3), hydrogen (H_2), nitrogen (N_2), and oxygen (O_2), depending on the feedstock and the digestion conditions.
- Biogas can be used as a renewable source of energy for heating, cooking, electricity generation, or vehicle fuel.
- Biogas can also reduce greenhouse gas emissions by capturing methane that would otherwise be released to the atmosphere from organic waste decomposition.

Production from organic waste materials and their environmental impact on society

- Organic waste materials are any biodegradable substances that come from plants or animals, such as food scraps, yard trimmings, paper, wood, leather, etc.
- Organic waste materials can be used for various purposes, such as composting, biogas production, biofertilizers, animal feed, etc.
- Composting is the process of breaking down organic waste materials into a nutrient-rich soil amendment that can improve soil quality and plant growth.
- Biogas production is the process of converting organic waste materials into a mixture of methane and carbon dioxide that can be used as a renewable energy source for heating, cooking, electricity generation, etc.
- Biofertilizers are organic substances that contain living microorganisms that can enhance the availability of nutrients and the biological activity of the soil.
- Animal feed is the food given to livestock, poultry, fish, etc. Organic waste materials can be used as a source of protein, carbohydrates, fats, vitamins, minerals, etc for animal feed.
- The environmental impact of organic waste materials depends on how they are managed and utilized. If organic waste materials are disposed of in landfills or dumps, they can cause several problems, such as :
 - Methane emissions: Organic waste materials decompose anaerobi-

cally (without oxygen) in landfills or dumps, producing methane, a potent greenhouse gas that contributes to global warming and climate change. Methane has a global warming potential 85 times higher than carbon dioxide over a 20-year period.

- Leachate contamination: Organic waste materials can release leachate, a liquid that contains harmful substances, such as heavy metals, pathogens, organic pollutants, etc. Leachate can pollute groundwater, soil, surface water, and air, posing risks to human and environmental health.
- Odor nuisance: Organic waste materials can emit unpleasant odors that can affect the quality of life of nearby residents and communities.
- Fire hazard: Organic waste materials can catch fire due to spontaneous combustion, electrical faults, or human negligence, causing air pollution, property damage, and health hazards.
- Disease transmission: Organic waste materials can attract rodents, insects, birds, and other animals that can carry and spread diseases, such as salmonella, typhoid, cholera, etc.
- On the other hand, if organic waste materials are used for composting, biogas production, biofertilizers, animal feed, etc, they can have several benefits, such as :
 - Greenhouse gas reduction: Organic waste materials can reduce greenhouse gas emissions by avoiding methane production in landfills or dumps, and by replacing fossil fuels with biogas or biofertilizers.
 - Resource recovery: Organic waste materials can recover valuable resources, such as nutrients, energy, organic matter, etc, that can be used for agricultural, industrial, or domestic purposes.
 - Waste minimization: Organic waste materials can reduce the amount of waste that needs to be disposed of, saving landfill space, transportation costs, and disposal fees.
 - Soil improvement: Organic waste materials can improve the physical, chemical, and biological properties of the soil, such as water retention, nutrient availability, soil structure, microbial activity, etc.
 - Crop productivity: Organic waste materials can enhance the growth and yield of crops, by providing organic matter, nutrients, and beneficial microorganisms to the plants.
 - Animal nutrition: Organic waste materials can provide a cheap and sustainable source of animal feed, by supplying protein, carbohydrates, fats, vitamins, minerals, etc to the animals.
- Therefore, organic waste materials can have a significant impact on the environment and society, depending on how they are managed and utilized. Proper management and utilization of organic waste materials can reduce waste generation, greenhouse gas emissions, and environmental pollution, while increasing resource recovery, soil quality, and crop productivity.

Unit-5: 8

- This unit covers the topic of **8-bit microprocessors**, which are electronic devices that can perform arithmetic, logic, and control operations on 8-bit binary data.
- An 8-bit microprocessor has an **8-bit data bus**, which means it can transfer 8 bits of data at a time between the processor and the memory or input/output devices.
- An 8-bit microprocessor also has an **8-bit address bus**, which means it can access up to $2^8 = 256$ memory locations or input/output ports.
- The most common 8-bit microprocessor is the **Intel 8085**, which was introduced in 1976 and used in many personal computers, calculators, and embedded systems.
- The Intel 8085 has the following features:
 - It has a **clock speed** of 3 MHz, which means it can execute 3 million instructions per second.
 - It has a **register set** of seven 8-bit general-purpose registers (A, B, C, D, E, H, and L), one 8-bit accumulator (A), one 8-bit flag register (F), and one 16-bit program counter (PC) and stack pointer (SP).
 - It has a **memory organization** of 64 KB of addressable memory, divided into 256 pages of 256 bytes each. The memory can be either RAM or ROM, or a combination of both.
 - It has a **instruction set** of 74 instructions, divided into five groups: data transfer, arithmetic, logical, branch, and control. The instructions can operate on 8-bit or 16-bit data, and can use immediate, register, direct, or indirect addressing modes.
 - It has a **interrupt system** of five hardware interrupts (TRAP, RST 7.5, RST 6.5, RST 5.5, and INTR) and one software interrupt (RST 0 to RST 7). The interrupts can be enabled or disabled by the processor, and can be used for external or internal events.
 - It has a **serial input/output** port (SID and SOD) that can be used for asynchronous data communication with other devices. The serial port can be controlled by the processor or by an external device.
 - It has a **parallel input/output** port (I/O/M, S0, S1, ALE, RD, and WR) that can be used for synchronous data communication with memory or input/output devices. The parallel port can be multiplexed or demultiplexed by the processor or by an external device.

Materials Chemistry:

- Materials chemistry is a branch of chemistry that deals with the design, synthesis, characterization, processing and understanding of materials with interesting or useful properties .
- Materials can be classified into different types based on their composition, structure, properties and applications. Some common types of materials are metals, ceramics, polymers, composites, biomaterials, nanomaterials,

etc.

- Materials chemistry aims to create new materials or improve existing ones by manipulating their molecular or atomic structures, using chemical reactions, physical methods or biological processes.
- Materials chemistry is an interdisciplinary field that draws from various areas of chemistry, such as organic, inorganic, physical, analytical, and biochemistry, as well as from other sciences, such as physics, engineering, biology, and mathematics.
- Materials chemistry has many applications in various fields, such as energy, electronics, medicine, catalysis, sensors, optics, magnetism, etc. Some examples of materials chemistry products are solar cells, batteries, LEDs, superconductors, drug delivery systems, catalysts, nanowires, etc.

Polymers; Classification, Polymerization processes, Thermosetting and Thermoplastic

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds.
- Polymers can be classified based on various factors, such as their source, structure, mode of synthesis, molecular forces, and thermal behavior.
- Based on the source, polymers can be natural (derived from plants or animals) or synthetic (made by chemical synthesis).
- Based on the structure, polymers can be linear (consisting of a single chain of monomers), branched (having side chains attached to the main chain), or cross-linked (having covalent bonds between different chains).
- Based on the mode of synthesis, polymers can be addition (formed by the addition of monomers without the elimination of any by-product) or condensation (formed by the elimination of a small molecule, such as water, from the monomers).
- Based on the molecular forces, polymers can be elastomers (having weak intermolecular forces and high elasticity), fibres (having strong intermolecular forces and high tensile strength), or plastics (having intermediate intermolecular forces and moderate properties).
- Based on the thermal behavior, polymers can be thermoplastic or thermosetting.
- Thermoplastic polymers are those that can be repeatedly softened by heating and then solidified by cooling, without altering their chemical structure or properties. They are usually linear or slightly branched polymers that can slide past each other when heated. Examples of thermoplastic polymers are polyethylene, polypropylene, polystyrene, and nylon.
- Thermosetting polymers are those that undergo a permanent chemical change when heated, forming a rigid and insoluble network of cross-linked chains. They cannot be remolded or recycled once cured. Examples of thermosetting polymers are bakelite, epoxy, and vulcanized rubber.
- Polymerization is the process of forming polymers from monomers. There are two main types of polymerization processes: chain polymerization and

step polymerization.

- Chain polymerization, also known as chain-growth polymerization, is a process in which the monomers are added one by one to the growing chain, initiated by a free radical, a cation, or an anion. The chain growth is terminated by a combination or disproportionation reaction. Chain polymerization is usually used for synthesizing addition polymers.
- Step polymerization, also known as step-growth polymerization, is a process in which the monomers react with each other in a stepwise manner, forming dimers, trimers, and so on, until a high molecular weight polymer is obtained. The monomers usually have two or more functional groups that can react with each other. Step polymerization is usually used for synthesizing condensation polymers.

Polymers, Polymer Blends and Composites, Conducting and Biodegradable polymers

- Polymers are large molecules composed of repeating units called monomers, which are linked by covalent bonds. Polymers can have various properties depending on their chemical structure, molecular weight, degree of crystallinity, and crosslinking.
- Polymer blends are mixtures of two or more polymers that are physically mixed but not chemically bonded. Polymer blends can improve the properties of individual polymers by combining their advantages and reducing their disadvantages. For example, polymer blends can enhance the mechanical strength, thermal stability, biocompatibility, or processability of polymers.
- Polymer composites are materials that consist of a polymer matrix and a reinforcing phase, such as fibers, particles, or nanomaterials. Polymer composites can have superior properties than the polymer matrix alone, such as high stiffness, strength, toughness, electrical conductivity, or thermal conductivity. Polymer composites are widely used in various applications, such as aerospace, automotive, biomedical, and energy.
- Conducting polymers are a class of polymers that can conduct electricity due to the presence of conjugated double bonds or delocalized electrons along the polymer backbone. Conducting polymers can have tunable electrical, optical, magnetic, and electrochemical properties by doping, blending, or forming composites with other materials. Conducting polymers have many potential applications in biosensors, tissue engineering, drug delivery, and antibacterial activity.
- Biodegradable polymers are polymers that can be degraded by biological agents, such as microorganisms or enzymes, into harmless products, such as water, carbon dioxide, or biomass. Biodegradable polymers can reduce the environmental impact of plastic waste and can be used for biomedical purposes, such as implants, sutures, or drug delivery systems. Biodegrad-

able polymers can have hydrolyzable linkages, such as ester, amide, or anhydride bonds, along the polymer chain that are susceptible to cleavage by water or enzymes.

Preparation, properties, industrial applications of Teflon, Lucite, Bakelite, Kelvar, Dacron

Teflon

- Teflon is a synthetic polymer of tetrafluoroethylene (C_2F_4) with the formula $(\text{C}_2\text{F}_4)_n$.
- It is prepared by free radical polymerization of tetrafluoroethylene in a pressurized vessel at high temperature.
- It has the following properties:
 - It is a white solid compound at room temperature with a density of about 2.2 g/cm^3 .
 - It has a high melting point of about 600 K and a low glass transition temperature of about 130 K .
 - It is chemically inert and resistant to most acids, bases, solvents, and oxidizing agents. The only chemicals that can affect it are alkali metals and some fluorinating agents.
 - It has a low coefficient of friction and a high non-stick property, making it suitable for coating surfaces that require lubrication or easy cleaning.
 - It has a low water absorption capacity and a high dielectric strength, making it useful for electrical insulation and waterproofing.
- It has the following industrial applications:
 - It is widely used as a coating material for cookware, bakeware, irons, hair dryers, and other appliances that require non-stick and heat-resistant surfaces.
 - It is also used as a coating material for pipes, valves, seals, gaskets, bearings, and other components that require corrosion resistance and low friction.
 - It is also used as a fabric protector for clothing, carpets, upholstery, and other textiles that require stain resistance and durability.
 - It is also used as a raw material for producing other fluoropolymers, such as PTFE, FEP, PFA, and ETFE, which have various applications in aerospace, chemical, medical, and electrical industries.

Lucite

- Lucite is a trade name for polymethyl methacrylate (PMMA), a transparent thermoplastic polymer of methyl methacrylate (MMA) with the formula $(\text{C}_5\text{O}_2\text{H}_8)_n$.
- It is prepared by free radical polymerization of methyl methacrylate in

the presence of an initiator, such as benzoyl peroxide, and a chain transfer agent, such as dodecyl mercaptan.

- It has the following properties:
 - It is a clear, colorless, and brittle solid at room temperature with a density of about 1.18 g/cm³.
 - It has a high glass transition temperature of about 378 K and a low melting point of about 463 K.
 - It is resistant to weathering, UV radiation, and most chemicals, except for strong acids, bases, and organic solvents.
 - It has a high refractive index of about 1.49 and a low birefringence, making it suitable for optical applications.
 - It has a high tensile strength, modulus of elasticity, and hardness, but a low impact strength and elongation at break, making it prone to cracking and shattering.
- It has the following industrial applications:
 - It is widely used as a substitute for glass in windows, lenses, screens, signs, displays, and other transparent or translucent objects that require clarity, light transmission, and shatter resistance.
 - It is also used as a molding material for making various objects, such as jewelry, toys, furniture, art, and dental prosthetics, that require aesthetic appeal, durability, and biocompatibility.
 - It is also used as a coating material for enhancing the gloss, scratch resistance, and UV protection of other substrates, such as wood, metal, and paper.

Bakelite

- Bakelite is a trade name for phenol-formaldehyde resin, a thermosetting polymer of phenol and formaldehyde with the formula (C₆H₆O-CH₂O)_n.
- It is prepared by condensation polymerization of phenol and formaldehyde in the presence of an acid or a base catalyst, such as hydrochloric acid, zinc chloride, or ammonia.
- It has the following properties :
 - It is a dark brown, hard, and brittle solid at room temperature with a density of about 1.25 g/cm³.
 - It has a high decomposition temperature of about 773 K and a low softening temperature of about 473

Thiokol, Nylon, Buna-N and Buna-S and their environmental impact on society, Speciality

- Thiokol is a synthetic rubber made from the co-polymerization of ethylene dichloride and sodium polysulfide. It is used as a sealant, adhesive, and binder for rocket propellants, gaskets, and coatings. It is also known as polysulfide rubber or Thiokol rubber.

- Nylon is a synthetic polymer made from the condensation of diamines and dicarboxylic acids. It is used as a fiber for textiles, carpets, ropes, and parachutes. It is also used as a plastic for gears, bearings, and casings. It is the first entirely synthetic fiber invented by DuPont in 1938.
- Buna-N is a synthetic rubber made from the co-polymerization of acrylonitrile and butadiene. It is also known as nitrile rubber or NBR. It is used as a seal, hose, and glove material for oil, fuel, and chemical resistance. It is also used as a medical and dental material for gloves and tubing.
- Buna-S is a synthetic rubber made from the co-polymerization of styrene and butadiene. It is also known as styrene butadiene rubber or SBR. It is used as a tire, conveyor belt, and shoe sole material for abrasion and wear resistance. It is also used as a latex substitute for paints and adhesives.

Environmental impact

- Thiokol is a non-biodegradable material that can persist in the environment for a long time. It can also release toxic gases such as hydrogen sulfide and sulfur dioxide when burned or decomposed. It can pose a threat to human health and wildlife if not properly disposed of or recycled.
- Nylon is a non-biodegradable material that can accumulate in landfills and oceans. It can also release greenhouse gases such as nitrous oxide and carbon dioxide when produced or incinerated. It can contribute to global warming and climate change if not reduced or reused.
- Buna-N is a non-biodegradable material that can leach harmful chemicals such as acrylonitrile and butadiene into the soil and water. It can also emit carcinogenic substances such as benzene and 1,3-butadiene when heated or exposed to sunlight. It can cause cancer and other diseases if not handled or stored safely.
- Buna-S is a non-biodegradable material that can degrade into microplastics and pollute the environment. It can also release volatile organic compounds such as styrene and butadiene into the air. It can cause respiratory and neurological problems if not controlled or ventilated.

Speciality

- Thiokol is a special synthetic rubber that has excellent elasticity, flexibility, and resistance to low temperatures, oxidation, and ozone. It can withstand extreme conditions and stresses without cracking or deteriorating. It is ideal for aerospace and marine applications.
- Nylon is a special synthetic polymer that has high strength, durability, and resistance to abrasion, heat, and chemicals. It can be spun into fine filaments or molded into various shapes and sizes. It is versatile and adaptable for various industrial and consumer applications.
- Buna-N is a special synthetic rubber that has high resistance to oil, fuel, and other chemicals. It can swell or shrink depending on the type and concentration of the solvent. It is selective and specific for different appli-

cations that require chemical compatibility and stability.

- Buna-S is a special synthetic rubber that has high resistance to abrasion, wear, and aging. It can be blended with other rubbers or fillers to modify its properties and performance. It is flexible and customizable for different applications that require mechanical strength and durability.

Polymers

- Polymers are any class of natural or synthetic substances composed of very large molecules, called macromolecules, that are multiples of simpler chemical units called monomers.
- Examples of polymers are rubber, plastics, and nylon.
- Polymers have different physical and chemical properties, which are affected by the structure, type of monomer units from which polymers are formed, and other factors.
- Some of the properties of polymers are:
 - Density: The mass per unit volume of a polymer. It depends on the molecular weight, degree of branching, and crystallinity of the polymer. Polymers with higher density are stronger and more rigid than those with lower density.
 - Melting point: The temperature at which a polymer changes from a solid to a liquid state. It depends on the intermolecular forces, degree of crystallinity, and molecular weight of the polymer. Polymers with higher melting point are more thermally stable and resistant to deformation than those with lower melting point.
 - Glass transition temperature: The temperature at which a polymer changes from a glassy (rigid and brittle) state to a rubbery (flexible and elastic) state. It depends on the flexibility, polarity, and molecular weight of the polymer. Polymers with higher glass transition temperature are more rigid and brittle than those with lower glass transition temperature.
 - Tensile strength: The maximum stress that a polymer can withstand before breaking. It depends on the molecular orientation, degree of cross-linking, and crystallinity of the polymer. Polymers with higher tensile strength are more durable and resistant to fracture than those with lower tensile strength.
 - Elongation: The percentage increase in length of a polymer when subjected to a tensile stress. It depends on the molecular orientation, degree of cross-linking, and crystallinity of the polymer. Polymers with higher elongation are more ductile and flexible than those with lower elongation.
 - Modulus: The ratio of stress to strain of a polymer. It measures the stiffness or elasticity of a polymer. It depends on the molecular orientation, degree of cross-linking, and crystallinity of the polymer. Polymers with higher modulus are more stiff and elastic than those

with lower modulus.

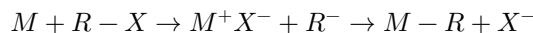
- Polymers can be classified into different types based on the structure, composition, and properties of the monomer units and the polymer chains. Some of the types of polymers are:
 - Homopolymers: Polymers that are made of only one type of monomer unit. For example, polyethylene, polypropylene, and polystyrene are homopolymers of ethylene, propylene, and styrene monomers, respectively.
 - Copolymers: Polymers that are made of two or more different types of monomer units. For example, nylon, polyester, and rubber are copolymers of different monomers.
 - Linear polymers: Polymers that have long and straight chains of monomer units. For example, polyethylene, polypropylene, and nylon are linear polymers.
 - Branched polymers: Polymers that have chains of monomer units with some branches or side chains. For example, low-density polyethylene, amylopectin, and glycogen are branched polymers.
 - Cross-linked polymers: Polymers that have chains of monomer units that are linked together by covalent bonds or other forces. For example, vulcanized rubber, bakelite, and epoxy resin are cross-linked polymers.
 - Thermoplastics: Polymers that can be melted and reshaped by heating and cooling. For example, polyethylene, polypropylene, and polystyrene are thermoplastics.
 - Thermosets: Polymers that cannot be melted and reshaped by heating and cooling. They undergo irreversible chemical changes when heated. For example, vulcanized rubber, bakelite, and epoxy resin are thermosets.

Organometallic Compounds: General methods of preparation and applications

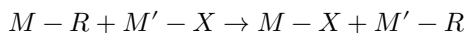
- Organometallic compounds are compounds that contain one or more metal-carbon bonds.
- They are classified according to the number and type of metal-carbon bonds, such as ionic, covalent, or mixed.
- They have diverse physical and chemical properties, depending on the nature of the metal and the organic ligand.

General methods of preparation

- Most organometallic compounds can be synthesized by using one of four M-C bond forming reactions :
 - Reaction with metal and transmetallation: The net reaction of an electropositive metal M and a halogen-substituted hydrocarbon is

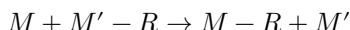


where R is an organic group and X is a halogen. This reaction can also be used to exchange one metal for another, such as

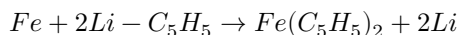


where M and M' are different metals.

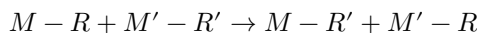
- Metal displacement: The reaction of a less electropositive metal M with an organometallic compound of a more electropositive metal M' is



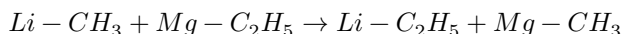
This reaction can be used to prepare organometallic compounds of transition metals, such as



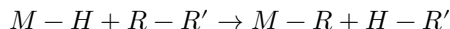
- Metathesis: The reaction of two organometallic compounds with different metals is



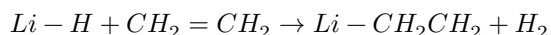
This reaction can be used to prepare organometallic compounds with different organic groups, such as



- Hydrometallation: The reaction of a metal hydride M-H with an unsaturated organic compound is



where R-R' is a double or triple bond. This reaction can be used to prepare organometallic compounds with alkenyl or alkynyl groups, such as



General applications

- Organometallic compounds have many applications in the field of chemistry. Some of them are listed below:
 - Organometallic compounds are used as homogeneous catalysts in certain commercial chemical reactions, such as hydroformylation, hydrogenation, polymerization, and metathesis.
 - They can be used as stoichiometric reagents for both research-oriented and industrial chemical reactions, such as cross-coupling, C-H activation, and carbonylation.
 - They can be used as precursors for the synthesis of other organometallic compounds, such as metal carbonyls, metal clusters, and metal-organic frameworks.

- They can be used as intermediates for the synthesis of organic compounds, such as alcohols, aldehydes, ketones, and carboxylic acids.
- They can be used as probes for the study of the structure and bonding of metal complexes, such as NMR, IR, and X-ray spectroscopy.

Organometallic Compounds (RMgX and LiAlH₄)

- Organometallic compounds are compounds that contain a metal-carbon bond, R-M.
- RMgX and LiAlH₄ are two important types of organometallic compounds that are widely used in organic synthesis.
- RMgX is also known as a Grignard reagent, named after Victor Grignard, who discovered them in 1900. X is a halogen atom (Cl, Br, I) and R is an alkyl or aryl group.
- LiAlH₄ is also known as lithium aluminium hydride, a powerful reducing agent that can reduce various carbonyl groups to alcohols.

Properties and Preparation of RMgX

- RMgX are usually prepared by reacting an alkyl or aryl halide with magnesium metal in an ether solvent, such as diethyl ether or tetrahydrofuran (THF) .
- RMgX are highly reactive and must be kept away from moisture, oxygen, and acidic compounds, as they can react violently with them .
- RMgX exist as tetrahedral complexes with the ether solvent molecules, as shown below for methylmagnesium chloride:

Methylmagnesium chloride complex

- RMgX are strong nucleophiles and bases, and can attack electrophilic carbon atoms, such as those in carbonyl groups, epoxides, alkyl halides, and carbon dioxide .
- RMgX can also react with other metal halides to form organometallic compounds of different metals, such as RLi, RCu, and RZn .

Properties and Preparation of LiAlH₄

- LiAlH₄ is usually prepared by reacting lithium hydride with aluminium chloride in an ether solvent.
- LiAlH₄ is a white solid that can ignite spontaneously in air, and must be stored under an inert atmosphere.
- LiAlH₄ is a very strong reducing agent that can reduce various carbonyl groups to alcohols, such as aldehydes, ketones, esters, amides, and carboxylic acids .
- LiAlH₄ can also reduce other functional groups, such as nitriles, imines, oximes, and nitro groups .

- LiAlH_4 reacts with water to produce hydrogen gas and lithium hydroxide, and with alcohols to produce alkoxides and aluminium alkoxides .

References

: https://chem.libretexts.org/Bookshelves/Inorganic_Chemistry/Chemistry_of_the_Main_Group_Elements
https://en.wikipedia.org/wiki/Grignard_reagent
<https://www.chem.ucalgary.ca/courses/351/Carey5th/Ch14/ch14-0.html>
<http://colapret.cm.utexas.edu/courses/Chapter%2015.pdf>

Course Outcomes:

- A course outcome is a statement that describes what a student should be able to do or demonstrate after completing a course.
- Course outcomes are usually derived from the course objectives, which are the broad goals or purposes of the course.
- Course outcomes should be specific, measurable, achievable, relevant, and time-bound (SMART).
- Course outcomes should align with the course content, activities, assessments, and learning outcomes of the program or degree.
- Course outcomes should be communicated to the students at the beginning of the course and throughout the course.
- Course outcomes should be evaluated and revised periodically based on feedback from students, instructors, and other stakeholders.

Upon completion of the course the student should be able to:

- Demonstrate an understanding of the basic concepts and principles of artificial intelligence, such as agents, search, knowledge representation, reasoning, planning, learning, natural language processing, computer vision, and robotics.
- Apply appropriate methods and techniques to solve various artificial intelligence problems, such as search algorithms, logic programming, probabilistic inference, machine learning algorithms, natural language processing tools, computer vision techniques, and robotics frameworks.
- Evaluate the strengths and limitations of different artificial intelligence approaches and compare their performance and applicability to different domains and scenarios.
- Design and implement artificial intelligence systems using various tools and platforms, such as Python, TensorFlow, PyTorch, OpenAI Gym, NLTK, OpenCV, ROS, etc.
- Communicate and present artificial intelligence concepts, methods, results, and applications effectively using oral, written, and visual means.

Units Course Outcomes Bloom's

- The course outcomes of units are the specific learning objectives that students are expected to achieve by the end of the course.
- The course outcomes of units are aligned with the Bloom's taxonomy of cognitive domains, which classifies learning into six levels: knowledge, comprehension, application, analysis, synthesis, and evaluation.
- The course outcomes of units are written in measurable and observable terms, using action verbs that correspond to the appropriate level of Bloom's taxonomy.
- The course outcomes of units are used to guide the design of the curriculum, the assessment methods, and the feedback strategies for the course.
- The course outcomes of units are communicated to the students at the beginning of the course, and are reviewed periodically throughout the course to monitor the students' progress and achievement.
- The course outcomes of units are evaluated at the end of the course, using various tools and techniques, such as rubrics, portfolios, tests, quizzes, projects, etc.
- The course outcomes of units are reviewed and revised regularly, based on the feedback from the students, the instructors, and the external stakeholders, such as employers, professional bodies, accreditation agencies, etc.

Level

- A level is a measure of the amount or degree of something, such as height, depth, quantity, quality, or intensity.
- A level can also refer to a position or rank in a hierarchy, such as a level of authority, education, or skill.
- A level can also mean a flat or horizontal surface or plane, such as the ground, a floor, or a table.
- A level can also be a tool or device for checking or adjusting the horizontal or vertical alignment of something, such as a spirit level or a laser level.
- A level can also be a stage or phase in a process, such as a level of difficulty, achievement, or development.

CO-1 Get an understanding of the theoretical principles of chemistry of molecular structure, bonding and

- Molecular structure refers to the arrangement of atoms in a molecule and the chemical bonds that hold them together.
- Bonding is the process of forming stable associations between atoms by sharing or transferring electrons.
- Theoretical principles of chemistry of molecular structure and bonding include:
 - The concept of atomic orbitals and molecular orbitals, which de-

scribe the regions of space where electrons are most likely to be found around atoms and molecules, respectively.

- The concept of hybridization, which explains how atomic orbitals can combine to form new hybrid orbitals with different shapes and orientations, suitable for forming bonds with other atoms.
- The concept of valence bond theory, which describes how covalent bonds are formed by the overlap of hybrid orbitals or atomic orbitals from different atoms, resulting in the formation of sigma and pi bonds.
- The concept of molecular orbital theory, which describes how covalent bonds are formed by the combination of atomic orbitals from different atoms to form molecular orbitals, which can be bonding or antibonding, depending on their energy and symmetry.
- The concept of Lewis structures, which represent the arrangement of valence electrons in a molecule using dots and lines, indicating the number and type of bonds and the formal charges of atoms.
- The concept of resonance, which explains how some molecules can have more than one valid Lewis structure, and how the actual structure is a hybrid of these resonance structures, with delocalized electrons and partial bonds.
- The concept of VSEPR theory, which predicts the three-dimensional shape of a molecule based on the repulsion between electron pairs around a central atom.
- The concept of polarity, which describes how some molecules have uneven distribution of charge, resulting in a dipole moment, which affects the intermolecular forces and physical properties of the molecule.
- The concept of electronegativity, which measures the tendency of an atom to attract electrons in a bond, and how it affects the bond polarity and the polarity of the molecule.
- The concept of bond order, bond length and bond energy, which describe the strength and stability of a bond, and how they are related to the number and type of bonds and the molecular orbital configuration.

Properties and Chemistry of Advanced Materials

Liquid Crystals

- Liquid crystals are substances that exhibit a phase of matter that has properties between those of conventional liquids and those of solid crystals.
- Liquid crystals can be classified into two types: thermotropic and lyotropic. Thermotropic liquid crystals change their phase in response to temperature, while lyotropic liquid crystals change their phase in response to the concentration of a solvent.

- Liquid crystals can also be classified based on the orientation and positional order of their molecules. The most common types are nematic, smectic, and cholesteric liquid crystals.
- Liquid crystals have various applications in display devices, sensors, optical fibers, and drug delivery systems.

Nanomaterials

- Nanomaterials are materials that have at least one dimension in the range of 1 to 100 nanometers. Nanomaterials can have unique physical, chemical, and biological properties due to their small size and large surface area.
- Nanomaterials can be classified into different types based on their shape, structure, and composition. Some examples are nanoparticles, nanotubes, nanowires, nanosheets, nanocomposites, and quantum dots.
- Nanomaterials can be synthesized by various methods, such as top-down, bottom-up, sol-gel, hydrothermal, and self-assembly techniques.
- Nanomaterials have various applications in electronics, energy, medicine, catalysis, and environmental remediation.

Graphite

- Graphite is an allotrope of carbon that has a layered structure of hexagonal rings of carbon atoms. Graphite is soft, slippery, and conducts electricity and heat well.
- Graphite can be intercalated with other atoms or molecules between its layers, which can change its properties. For example, graphite intercalated with potassium can become a superconductor.
- Graphite can also be exfoliated into thin layers of graphene, which is a two-dimensional material with exceptional mechanical, electrical, and optical properties.
- Graphite has various applications in pencils, lubricants, batteries, electrodes, and composite materials.

Fullerene

- Fullerene is an allotrope of carbon that has a spherical, cylindrical, or ellipsoidal shape. The most common type of fullerene is C₆₀, which has 60 carbon atoms arranged in a soccer ball-like structure.
- Fullerene can act as an electrophile and an oxidizing agent in chemical reactions, as it can accept electrons from other molecules. Fullerene can also be doped with metals to exhibit superconductivity.
- Fullerene can also form larger structures, such as nanotubes, nanorods, and nanohorns, which have different properties and applications.
- Fullerene has various applications in nanomedicine, solar cells, sensors, and materials science.

Principles of Green Chemistry

Green chemistry is the approach in chemical sciences that efficiently uses renewable raw materials, eliminating waste and avoiding the use of toxic and hazardous reagents and solvents in the manufacture and application of chemical products.

There are twelve basic principles of green chemistry that can be grouped into “Reducing Risk” and “Minimizing the Environmental Footprint”. They are:

1. **Prevention:** It is better to prevent waste than to treat or clean up waste after it has been created.
2. **Atom Economy:** Synthetic methods should be designed to maximize the incorporation of all materials used in the process into the final product.
3. **Less Hazardous Chemical Syntheses:** Wherever practicable, synthetic methods should be designed to use and generate substances that possess little or no toxicity to human health and the environment.
4. **Designing Safer Chemicals:** Chemical products should be designed to affect their desired function while minimizing their toxicity.
5. **Safer Solvents and Auxiliaries:** The use of auxiliary substances (e.g., solvents, separation agents, etc.) should be made unnecessary wherever possible and innocuous when used.
6. **Design for Energy Efficiency:** Energy requirements of chemical processes should be recognized for their environmental and economic impacts and should be minimized. If possible, synthetic methods should be conducted at ambient temperature and pressure.
7. **Use of Renewable Feedstocks:** A raw material or feedstock should be renewable rather than depleting whenever technically and economically practicable.
8. **Reduce Derivatives:** Unnecessary derivatization (use of blocking groups, protection/ deprotection, temporary modification of physical/chemical processes) should be minimized or avoided if possible, because such steps require additional reagents and can generate waste.
9. **Catalysis:** Catalytic reagents (as selective as possible) are superior to stoichiometric reagents.
10. **Design for Degradation:** Chemical products should be designed so that at the end of their function they break down into innocuous degradation products and do not persist in the environment.
11. **Real-time Analysis for Pollution Prevention:** Analytical methodologies need to be further developed to allow for real-time, in-process monitoring and control prior to the formation of hazardous substances.
12. **Inherently Safer Chemistry for Accident Prevention:** Substances and the form of a substance used in a chemical process should be chosen to minimize the potential for chemical accidents, including releases, explosions, and fires.

These principles aim to contribute to sustainable development by reducing the

environmental impact and health risks of chemical products and processes. They also offer economic benefits by improving the efficiency and competitiveness of the chemical industry.

K3

- K3 is a type of surface mount technology (SMT) component package that is used for integrated circuits (ICs).
- K3 stands for **Kilopass 3**, which is a trademark of Kilopass Technology Inc., a company that specializes in non-volatile memory (NVM) solutions.
- K3 packages have a **rectangular shape** with **32 pins** on each side, for a total of **128 pins**. The pins are arranged in a **0.5 mm pitch** grid, meaning the distance between the centers of adjacent pins is 0.5 mm.
- K3 packages have a **body size** of **11 mm x 11 mm** and a **height** of **1.2 mm**. The body is made of **plastic** and the pins are made of **copper**.
- K3 packages are designed for **high-density** and **low-power** applications, such as **smart cards**, **security chips**, **embedded systems**, and **IoT devices**.
- K3 packages are compatible with **standard SMT equipment** and **processes**, such as **reflow soldering**, **pick and place**, and **inspection**. They can be mounted on **printed circuit boards (PCBs)** using **solder paste** and **stencils**.
- K3 packages have a **temperature range** of **-40°C to +125°C** and a **voltage range** of **1.8 V to 3.6 V**. They can store up to **4 Mb** of data in **one-time programmable (OTP)** or **multi-time programmable (MTP)** modes.
- K3 packages have a **high reliability** and **long endurance**, with a **data retention** of **10 years** and a **write/erase cycle** of **10,000 times**. They also have a **low standby current** of **1 μ A** and a **fast access time** of **50 ns**.

CO-2 Apply the fundamental concepts of determination of structure with various spectral techniques

- Spectroscopy is the study of how electromagnetic radiation, across a spectrum of different wavelengths, interacts with molecules - and how these interactions can be quantified, analyzed, and ultimately interpreted to gain information about molecular structure.
- The most important spectroscopic techniques for structure determination are ultraviolet and visible spectroscopy, infrared spectroscopy, and nuclear magnetic resonance spectroscopy.
- Ultraviolet and visible spectroscopy measures the absorption of light by molecules in the range of 200-800 nm. This absorption corresponds to the excitation of electrons from lower to higher energy levels. The absorption spectrum can provide information about the presence of conjugated pi systems, chromophores, and functional groups in the molecule.

- Infrared spectroscopy measures the absorption of light by molecules in the range of 2500-25,000 nm. This absorption corresponds to the vibration of bonds and groups in the molecule. The infrared spectrum can provide information about the types and numbers of bonds and functional groups in the molecule, as well as their relative positions and orientations.
- Nuclear magnetic resonance spectroscopy measures the absorption of radiofrequency waves by nuclei in a magnetic field. This absorption corresponds to the spin transitions of nuclei from lower to higher energy levels. The nuclear magnetic resonance spectrum can provide information about the number and types of atoms in the molecule, their relative positions and environments, and their connectivity and stereochemistry.
- Each spectroscopic technique has its own advantages and limitations, and they can be combined to give complementary and overlapping information about the structure of a molecule. For example, ultraviolet and visible spectroscopy can identify the presence of a benzene ring, but cannot distinguish between different isomers of benzene. Infrared spectroscopy can identify the presence of a carbonyl group, but cannot distinguish between different types of carbonyl compounds. Nuclear magnetic resonance spectroscopy can identify the number and types of carbon and hydrogen atoms, but cannot detect atoms with zero or even nuclear spin, such as oxygen and nitrogen.
- To apply the fundamental concepts of determination of structure with various spectral techniques, one should follow a systematic approach that involves the following steps:
 - Obtain and analyze the molecular formula of the compound, if available, and calculate the degrees of unsaturation.
 - Obtain and analyze the ultraviolet and visible spectrum of the compound, if available, and identify the presence of conjugated pi systems, chromophores, and functional groups.
 - Obtain and analyze the infrared spectrum of the compound, and identify the types and numbers of bonds and functional groups.
 - Obtain and analyze the nuclear magnetic resonance spectrum of the compound, and identify the number and types of atoms, their relative positions and environments, and their connectivity and stereochemistry.
 - Compare and integrate the information from different spectroscopic techniques, and propose a possible structure for the compound that is consistent with all the data.
 - Verify the proposed structure by comparing it with known structures or by performing additional experiments, such as mass spectrometry, X-ray crystallography, or electron microscopy.

Stereochemistry

Stereochemistry is the branch of chemistry that deals with the three-dimensional arrangement of atoms and molecules and the effect of this on chemical reactions . Stereochemistry focuses on the relationships between stereoisomers, which are molecules that have the same molecular formula and sequence of bonded atoms, but differ in the orientation of their atoms in space.

Some of the main topics in stereochemistry are:

- **Chirality:** the property of a molecule that makes it non-superimposable on its mirror image. A molecule that has chirality is called a chiral molecule, and it can exist in two forms called enantiomers, which are mirror images of each other. Enantiomers have the same physical and chemical properties, except for their interaction with polarized light and other chiral molecules .
- **Optical activity:** the ability of a chiral molecule to rotate the plane of polarized light. The direction and degree of rotation depend on the structure and configuration of the molecule, as well as the wavelength and temperature of the light. The specific rotation of a molecule is a characteristic physical property that can be used to identify and quantify it .
- **Configuration:** the spatial arrangement of the atoms or groups in a molecule, which can be determined by using conventions such as the Cahn-Ingold-Prelog (CIP) rules and the R/S system for chiral centers, and the E/Z system for alkenes. Configuration is usually fixed by the presence of double bonds or chiral centers, and can only be changed by breaking and forming bonds .
- **Conformation:** the different spatial arrangements of the atoms or groups in a molecule that result from rotation around single bonds. Conformation can be influenced by factors such as steric strain, torsional strain, and hydrogen bonding. Conformation can be represented by using models such as the Newman projection and the Fischer projection .
- **Stereochemical reactions:** the chemical reactions that involve changes in the stereochemistry of the reactants or products, such as inversion, retention, racemization, and resolution. Stereochemical reactions can be influenced by the mechanism, the solvent, the catalyst, and the stereochemistry of the starting materials .

Stereochemistry is an important aspect of organic chemistry, biochemistry, pharmacology, and materials science, as it affects the structure, function, and interactions of molecules. Stereochemistry can also be used to synthesize and characterize new compounds, and to understand the mechanisms and kinetics of chemical reactions .

K4

- K4 is a **symmetric-key algorithm** that was used by the German Navy during World War II for encrypting messages.
- K4 is also known as **Tirpitz cipher** or **Offizier** (officer) cipher, as it was mainly used by high-ranking naval officers.
- K4 is based on the **Enigma machine**, but with an additional **fourth rotor** that increased the security of the encryption.
- K4 has a **plugboard** that allows the user to swap pairs of letters before and after the rotors, adding another layer of complexity to the cipher.
- K4 has a **key** that consists of three parts: the **rotor settings**, the **ring settings**, and the **plugboard settings**. The rotor settings determine the order and position of the four rotors, the ring settings determine the offset of the alphabet rings on each rotor, and the plugboard settings determine the pairs of letters that are swapped.
- K4 has a **turnover mechanism** that causes the rotors to rotate by one step after each letter is encrypted or decrypted. The rightmost rotor rotates every time, the second rotor rotates after the rightmost rotor completes a full turn, the third rotor rotates after the second rotor completes a full turn, and the fourth rotor rotates only when the user presses a special key on the keyboard.
- K4 is a **polyalphabetic substitution cipher**, meaning that each letter is replaced by another letter according to a changing alphabet. The alphabet changes depending on the position and settings of the rotors and the plugboard.
- K4 is a **reciprocal cipher**, meaning that the same machine and key can be used for both encryption and decryption. To encrypt a message, the user types the plaintext on the keyboard and notes the ciphertext that appears on the lampboard. To decrypt a message, the user types the ciphertext on the keyboard and notes the plaintext that appears on the lampboard.
- K4 is a **strong cipher**, meaning that it is very difficult to break without knowing the key or having access to the machine. The number of possible keys for K4 is estimated to be around 10^{23} , which is much larger than the number of keys for the standard Enigma machine (around 10^{20}).
- K4 was **broken** by the Allied codebreakers at Bletchley Park, who used a combination of **cryptanalysis**, **intelligence**, and **special devices** to crack the messages. Some of the devices they used were the **Bombe**, an electromechanical machine that simulated multiple Enigma machines and searched for possible keys, and the **Colossus**, an early electronic computer that helped to break the more complex Lorenz cipher.

CO-3 Utilize the theory of construction of electrodes, batteries and fuel cells in redesigning new engineering

- Electrodes are conductors that allow electric current to flow between a device and a solution or a gas. They are usually made of metals or carbon materials that have high electrical conductivity and chemical stability.
- Batteries are devices that convert chemical energy into electrical energy by means of electrochemical reactions. They consist of one or more cells, each containing a positive electrode (cathode), a negative electrode (anode), and an electrolyte that allows the flow of ions.
- Fuel cells are devices that generate electricity and heat by combining a fuel (such as hydrogen) and an oxidant (such as oxygen) across an electrolyte. They are similar to batteries, but they do not run down or need recharging as long as the fuel and oxidant are supplied. They also have two electrodes, an anode and a cathode, where the fuel is oxidized and the oxidant is reduced.
- The theory of construction of electrodes, batteries and fuel cells is based on the principles of electrochemistry, thermodynamics, kinetics, and materials science. It involves the design and optimization of the electrode materials, the electrolyte composition, the cell configuration, and the operating conditions to achieve the desired performance, efficiency, durability, and cost.
- The theory of construction of electrodes, batteries and fuel cells can be utilized in redesigning new engineering applications, such as:
 - Developing new electrode materials with higher catalytic activity, selectivity, stability, and conductivity for various electrochemical reactions, such as water splitting, carbon dioxide reduction, oxygen reduction, and hydrogen oxidation.
 - Improving the electrolyte properties, such as ionic conductivity, stability, compatibility, and safety, by using different types of electrolytes, such as aqueous, non-aqueous, solid, liquid, gel, or polymer electrolytes.
 - Designing new cell architectures, such as bipolar, monopolar, tubular, planar, or stacked cells, to enhance the mass and charge transport, heat dissipation, and power density of the devices.
 - Optimizing the operating conditions, such as temperature, pressure, humidity, flow rate, and load, to maximize the efficiency, durability, and reliability of the devices.
 - Integrating the devices with renewable energy sources, such as solar, wind, or hydro power, to store and utilize the excess energy and reduce the greenhouse gas emissions.
 - Developing new applications, such as electric vehicles, portable electronics, backup power, distributed generation, and cogeneration, that can benefit from the advantages of the devices, such as high energy density, low emissions, and modularity.

Products and Categorize the Reasons for Corrosion and Study Methods to Control Corrosion and Develop

- Corrosion is the deterioration of metals due to chemical reactions with the environment, which result in functional failure of components.
- Corrosion can be classified into different types based on the appearance, mechanism, or environment of the metal. Some common types are:
 - Uniform corrosion: The most common type of corrosion, where the metal surface is uniformly attacked by the corrosive medium. The rate of corrosion depends on the nature of the metal and the environment.
 - Localized corrosion: A type of corrosion that occurs at discrete sites on the metal surface, such as pits, crevices, or cracks. This type of corrosion can cause severe damage to the metal, even if the overall corrosion rate is low.
 - Galvanic corrosion: A type of corrosion that occurs when two dissimilar metals are in contact with each other and an electrolyte, such as water or soil. The more active metal (anode) corrodes faster than the less active metal (cathode), creating an electric current between them.
 - Stress corrosion cracking: A type of corrosion that occurs when a metal is subjected to tensile stress and a corrosive environment, causing cracks to form and propagate in the metal. This type of corrosion can lead to catastrophic failure of the metal, especially in high-strength alloys.
 - Intergranular corrosion: A type of corrosion that occurs along the grain boundaries of a metal, usually due to the presence of impurities or precipitates in the metal. This type of corrosion can weaken the metal and reduce its ductility.
- Corrosion can be prevented or controlled by various methods, such as:
 - Protective coatings: A layer of another material that is applied to the metal surface to isolate it from the corrosive environment. The coating can be metallic, such as zinc or chromium, or non-metallic, such as paint or plastic.
 - Cathodic protection: An electrochemical technique that applies a negative potential to the metal surface, making it the cathode and preventing it from corroding. This can be achieved by using a sacrificial anode, such as magnesium or zinc, or by using an external power source.
 - Corrosion inhibitors: Chemical substances that are added to the corrosive medium to reduce the rate of corrosion. They can act by forming a protective film on the metal surface, by inhibiting the anodic or cathodic reactions, or by scavenging the corrosive agents, such as oxygen or chloride.

- Corrosion-resistant alloys: Metals that are designed to have high resistance to corrosion in specific environments, such as stainless steel or titanium. They can have different compositions, microstructures, or surface treatments to enhance their corrosion performance.
- Corrosion management: A systematic approach that involves planning, designing, operating, and maintaining the critical assets to reduce the cost and risk of corrosion. It can include corrosion monitoring, inspection, testing, analysis, and mitigation.

Understanding of Chemistry of Engineering Materials (Cement)

Cement is a binding material that is used in construction and civil engineering. It is composed of various chemical compounds that react with water to form a hard and durable mass. The following points summarize the chemistry of cement:

- Cement can be classified into two main types: hydraulic and non-hydraulic. Hydraulic cement hardens due to the addition of water, while non-hydraulic cement hardens by carbonation with the air.
- The most common type of hydraulic cement is Portland cement, which is made up of four main compounds: tricalcium silicate ($3\text{CaO} \cdot \text{SiO}_2$), dicalcium silicate ($2\text{CaO} \cdot \text{SiO}_2$), tricalcium aluminate ($3\text{CaO} \cdot \text{Al}_2\text{O}_3$), and a tetra-calcium aluminoferrite ($4\text{CaO} \cdot \text{Al}_2\text{O}_3\text{Fe}_2\text{O}_3$).
- The chemical reactions that take place when water is added to Portland cement are called hydration. Hydration produces calcium silicate hydrate (C-S-H) and calcium hydroxide ($\text{Ca}(\text{OH})_2$), which are responsible for the strength and durability of the cement paste.
- The rate and extent of hydration depend on various factors, such as the composition, fineness, temperature, and water-cement ratio of the cement. Hydration can be divided into five stages: initial hydrolysis, induction, acceleration, deceleration, and steady state.
- Cement is usually mixed with inert materials, such as sand or crushed stone, to form mortar or concrete. The properties of the final product depend on the type and proportion of the cement and the aggregates, as well as the curing conditions and the environmental factors.

K3

- K3 is a type of synthetic cannabinoid that mimics the effects of THC, the main psychoactive component of cannabis.
- K3 is also known as Spice, K2, or synthetic marijuana, and is sold as a herbal incense or potpourri that can be smoked or vaporized.
- K3 is not detected by standard drug tests and is often marketed as a legal alternative to cannabis, but it is illegal in many countries and states due

to its health risks and potential for abuse.

- K3 can cause unpredictable and adverse effects such as anxiety, paranoia, hallucinations, psychosis, seizures, vomiting, chest pain, and even death.
- K3 can also lead to dependence and withdrawal symptoms such as insomnia, irritability, depression, and cravings.
- K3 is not safe to use and can have serious consequences for physical and mental health.

Hello, I am Sydney, your AI assistant. I can help you with any topic you want to learn or discuss. You have chosen the topic of CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of. Here is some content that you can use as study material for this topic:

CO-4 Develop understanding of the sources, impurities and hardness of water, apply the concepts of

- Sources of water: Water is a natural resource that is essential for life. Water can be obtained from various sources, such as surface water (rivers, lakes, ponds, etc.), groundwater (wells, springs, aquifers, etc.), rainwater (collected from roofs, tanks, etc.), seawater (desalinated by reverse osmosis, distillation, etc.), and wastewater (treated by biological, chemical, or physical methods).
- Impurities of water: Water is rarely pure in nature. It contains various impurities, such as dissolved solids (salts, minerals, metals, etc.), dissolved gases (oxygen, carbon dioxide, nitrogen, etc.), suspended solids (clay, silt, sand, etc.), colloidal particles (organic matter, bacteria, viruses, etc.), and dissolved organic compounds (pesticides, herbicides, pharmaceuticals, etc.). These impurities can affect the quality, taste, odor, color, and suitability of water for various purposes.
- Hardness of water: Hardness of water is a measure of the concentration of calcium and magnesium ions in water. These ions form insoluble compounds with soap, causing scum and reducing the lathering ability of soap. Hard water also causes scaling and corrosion of pipes, boilers, and appliances. Hardness of water can be expressed in various units, such as parts per million (ppm), milligrams per liter (mg/L), grains per gallon (gpg), or degrees of hardness (dH). Hardness of water can be classified as soft (< 60 ppm), moderately hard (60-120 ppm), hard (120-180 ppm), or very hard (> 180 ppm).
- Concepts of water treatment: Water treatment is the process of removing or reducing the impurities of water to make it suitable for various purposes, such as drinking, irrigation, industrial use, etc. Water treatment can involve various methods, such as coagulation and flocculation (adding chemicals to form larger particles that can be settled or filtered), sedimentation (allowing the heavier particles to settle at the bottom of a tank), filtration (passing water through a porous medium to remove suspended solids and colloids), disinfection (killing or inactivating microorganisms

by chlorine, ozone, ultraviolet light, etc.), softening (removing or replacing calcium and magnesium ions by ion exchange, lime-soda process, etc.), and membrane processes (separating water from dissolved solids by reverse osmosis, nanofiltration, ultrafiltration, etc.).

Determination of Calorific Values and Analysis of Coal

- Calorific value is a measure of the amount of energy produced from a unit weight of coal when it is combusted in oxygen.
- Calorific value can be determined by different methods, such as thermal analysis, bomb calorimetry, and laser-induced breakdown spectroscopy.
- Thermal analysis is a technique that measures the changes in physical and chemical properties of coal as a function of temperature.
 - Thermal analysis can be performed using different instruments, such as thermogravimetry (TG), differential thermal analysis (DTA), and differential scanning calorimetry (DSC).
 - Thermal analysis can provide information on the moisture, volatile matter, fixed carbon, and ash content of coal, as well as the combustion characteristics and kinetics.
 - Thermal analysis can be used to estimate the calorific value of coal by using empirical correlations or mathematical models based on the mass loss and heat flow data.
- Bomb calorimetry is a method that measures the heat released by the complete combustion of coal in a sealed vessel under constant volume and pressure.
 - Bomb calorimetry is performed using a device called a bomb calorimeter, which consists of a steel bomb, a water jacket, a thermometer, and a stirrer.
 - Bomb calorimetry can provide the gross calorific value of coal, which is the heat produced when the water vapor in the combustion products is condensed.
 - Bomb calorimetry can also provide the net calorific value of coal, which is the heat produced when the water vapor in the combustion products is not condensed, by subtracting the latent heat of vaporization of water from the gross calorific value.
 - Bomb calorimetry is considered the standard method for determining the calorific value of coal, and it is specified by the ASTM D5865-12 standard.
- Laser-induced breakdown spectroscopy (LIBS) is a method that measures the elemental composition of coal by analyzing the emission spectra of the plasma generated by a laser pulse on the coal surface.
 - LIBS can be coupled with acoustic levitation, which is a technique that suspends a small sample of coal in the air by using sound waves,

- to avoid the interference of the sample holder and the background noise.
- LIBS can provide information on the carbon, hydrogen, nitrogen, sulfur, and oxygen content of coal, which are related to the calorific value.
- LIBS can be used to estimate the calorific value of coal by using multivariate regression methods, such as partial least squares (PLS), principal component regression (PCR), or support vector regression (SVR), based on the spectral intensity data.

K3

- K3 is a synthetic, artificially produced form of vitamin K that doesn't occur naturally .
- K3 is also known as menadione .
- K3 may be converted into K2 in your liver.
- K3 is important for blood clotting and bone health.
- K3 has demonstrated anticancer and antibacterial properties in test-tube studies.
- K3 can also cause oxidative stress, cell damage, and liver toxicity .
- K3 is not approved for human use and is banned in many countries .
- K3 is sometimes used as a supplement for animals, especially poultry and fish .
- K3 is also the name of a schedule form for reporting a partner's share of income, deductions, credits, etc. from international sources in the U.S. tax system.

CO-5 Develop the understanding of Chemical structure of polymers and its effect on their various properties

- Polymers are very long carbon-based molecules that have a wide range of beneficial structural (and sometimes thermal) properties.
- The chemical structure of polymers refers to the arrangement of atoms and bonds in the polymer chains, as well as the degree of branching, cross-linking, and crystallinity.
- The chemical structure of polymers affects their physical properties, such as density, melting point, strength, elasticity, solubility, and reactivity.
- Some of the factors that influence the physical properties of polymers are:
 - Chain length: Longer polymer chains have more intermolecular forces, such as van der Waals forces and hydrogen bonding, which make them stronger and stiffer .
 - Side groups: The presence of different functional groups or atoms attached to the polymer chains can affect the polarity, flexibility, and reactivity of the polymers. For example, polyvinyl chloride (PVC)

has chlorine atoms as side groups, which make it more polar and less flexible than polyethylene.

- Branching: The degree of branching refers to the number of side chains that branch off from the main polymer chain. More branching reduces the ability of the polymer chains to pack closely together, which lowers the density, crystallinity, and melting point of the polymers. For example, low-density polyethylene (LDPE) has more branching than high-density polyethylene (HDPE), which makes it softer and more flexible .
 - Cross-linking: The degree of cross-linking refers to the number of covalent bonds that link different polymer chains together. More cross-linking increases the rigidity, strength, and thermal stability of the polymers, but also reduces their solubility and elasticity. For example, vulcanized rubber has more cross-linking than natural rubber, which makes it more durable and resistant to heat.
 - Crystallinity: The degree of crystallinity refers to the fraction of the polymer chains that are arranged in a regular and ordered manner. More crystallinity increases the density, strength, and melting point of the polymers, but also reduces their flexibility and transparency. For example, nylon has more crystallinity than polypropylene, which makes it stronger and more resistant to abrasion.
- The chemical structure of polymers can be modified by various methods, such as polymerization, copolymerization, blending, and chemical modification, to obtain desired physical properties for different applications

Polymers: Engineering Materials, Applications and Chemistry

Polymers are long chain macromolecules built from repeated smaller chemical structures or ‘-mers’. They are found widely in the natural world (e.g. proteins, rubber, cellulose) and, increasingly the past century, in synthetic form (e.g. “plastics”). Polymers have many attractive properties, such as monodispersity, biocompatibility, controlled composition and chain length, and tunable chemical properties. Polymers can also be endowed with functions by using certain special preparation methods or by introducing functional groups or fillers into material.

Some of the applications of polymers as engineering materials are:

- In aircraft, aerospace, and sports equipment: Polymers can be used to make lightweight, strong, and durable materials that can withstand high stress and temperature. Examples are carbon fiber reinforced polymers (CFRP), epoxy resins, and polyimides.
- In 3D printing plastics: Polymers can be used to create complex shapes

and structures with high precision and accuracy. Examples are acrylonitrile butadiene styrene (ABS), polylactic acid (PLA), and polyethylene terephthalate glycol (PETG).

- In biopolymers and biomolecules: Polymers can be used to mimic natural structures and functions in biological systems. Examples are DNA, proteins, polysaccharides, and lipids. They can also be used for molecular recognition, drug delivery, biosensor devices, tissue engineering, and cosmetics.
- In holography: Polymers can be used to create holograms that can store and display information in three dimensions. Examples are photopolymers, dichromated gelatin, and thermoplastic polymers.
- In water purification: Polymers can be used to remove contaminants and impurities from water. Examples are polyacrylamide, polyvinyl alcohol, and chitosan.
- In printed circuit board substrates: Polymers can be used to provide electrical insulation and mechanical support for electronic components. Examples are polyimide, epoxy, and polytetrafluoroethylene (PTFE).
- In green chemicals: Polymers and biopolymers can be used to replace petroleum-based chemicals and reduce environmental impact. Examples are polylactic acid (PLA), polyhydroxyalkanoates (PHA), and cellulose.

The chemistry of polymers involves the synthesis, characterization, and modification of polymer chains and structures. Some of the important concepts and methods in polymer chemistry are:

- Polymerization: The process of forming polymer chains from monomers. There are two main types of polymerization: chain-growth and step-growth. Chain-growth polymerization involves the addition of monomers to the end of a growing chain, while step-growth polymerization involves the condensation of monomers with the elimination of a small molecule.
- Copolymerization: The process of forming polymer chains from two or more different types of monomers. There are different types of copolymers, such as random, alternating, block, and graft copolymers, depending on the arrangement of the monomers along the chain.
- Crosslinking: The process of forming covalent bonds between polymer chains to create a network structure. Crosslinking can increase the strength, rigidity, and stability of polymers. Examples of crosslinked polymers are vulcanized rubber, bakelite, and epoxy.
- Polymer degradation: The process of breaking down polymer chains due to chemical, thermal, mechanical, or biological factors. Polymer degradation can affect the properties and performance of polymers. Examples of polymer degradation are oxidation, hydrolysis, photodegradation, and biodegradation.
- Polymer characterization: The process of determining the structure, composition, molecular weight, and properties of polymers. Some of the techniques used for polymer characterization are spectroscopy, chromatography, microscopy, rheology, and thermal analysis.

- Polymer modification: The process of changing the properties and functions of polymers by introducing chemical or physical changes. Some of the methods used for polymer modification are functionalization, blending, grafting, and doping.

Industrial Process

- An industrial process is a series of operations or steps that transform raw materials or inputs into finished products or outputs.
- Industrial processes can involve physical, chemical, biological, or mechanical changes, or a combination of these.
- Industrial processes can be classified into two main types: continuous and batch.
 - Continuous processes operate without interruption and produce a steady stream of output. Examples of continuous processes are oil refining, paper making, and electricity generation.
 - Batch processes operate in discrete cycles and produce a finite amount of output per cycle. Examples of batch processes are baking, brewing, and pharmaceutical manufacturing.
- Industrial processes can have various objectives, such as increasing efficiency, reducing costs, improving quality, enhancing safety, or minimizing environmental impact.
- Industrial processes can be optimized by applying various methods, such as lean manufacturing, six sigma, total quality management, or statistical process control.

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Reference Books:

- Reference books are books that contain factual information on various topics, such as dictionaries, encyclopedias, atlases, almanacs, directories,

handbooks, manuals, etc.

- Reference books are usually consulted for specific information, rather than read cover to cover.
- Reference books are often arranged alphabetically, chronologically, geographically, or by subject, depending on the type and purpose of the book.
- Reference books are usually kept in a separate section of a library, called the reference collection, and are not available for borrowing.
- Reference books are updated regularly to reflect the latest information and developments in various fields of knowledge.
- Reference books are useful sources for finding definitions, facts, statistics, background information, overviews, and bibliographies on a topic.

Engineering Chemistry by Rath & Singh, 2nd Edition, Cengage Learning India Pvt Ltd Delhi

- Engineering Chemistry is a textbook that covers all topics taught to the first-year undergraduate students of all engineering disciplines offered by various Indian universities .
- The book follows the syllabi of Biju Patnaik University of Technology (BPUT), KIIT (Deemed University), and SOA (Deemed University) in the state of Orissa.
- The book covers topics from diverse branches of chemistry, viz., general, inorganic, organic, and analytical chemistry in addition to a topic, nano-materials, from material science .
- The book aims to provide a comprehensive and systematic coverage of the theory behind the concepts and their applications in engineering.
- The book has a neat and student-friendly visual presentation with rich pedagogy to support and illustrate the material for easy and effective learning.
- The book has a variety of solved and unsolved problems/exercises to help reinforce learning and boost the confidence of the learner/reader.
- The book is suitable for students who want to learn the basics of chemistry and its applications in engineering, as well as for those who want to prepare for exams.

2. Engineering Chemistry by SS Dara, S Chand & Co Ltd

- Engineering Chemistry by SS Dara, S Chand & Co Ltd is a textbook for engineering students that covers the basic concepts and applications of chemistry in various fields of engineering.
- The book is divided into 16 chapters, each dealing with a specific topic such as water, fuels, corrosion, polymers, lubricants, electrochemistry, spectroscopy, etc.
- The book provides a comprehensive and systematic presentation of the theoretical and practical aspects of engineering chemistry, with numerous

examples, problems, and exercises.

- The book also includes appendices that contain useful data and tables, such as atomic weights, molecular weights, solubility products, electrode potentials, etc.
- The book is suitable for the first year engineering students of all branches, as well as for the students preparing for competitive examinations such as GATE, IES, etc.

3. Engineering Chemistry by Jain & Jain, S.Chand & Comp, New Delhi

- Engineering Chemistry by Jain & Jain is a textbook written for the students of all branches of engineering, covering the syllabus of various reputed universities throughout the country .
- The book is divided into 26 chapters, covering topics such as atomic structure, chemical bonding, phase rule, solutions, electrochemistry, corrosion, polymers, fuels, water, instrumental methods of analysis, environmental chemistry, nanomaterials, and green chemistry.
- The book aims to provide a clear and concise presentation of the concepts, principles, and applications of engineering chemistry, with an emphasis on problem-solving and numerical techniques.
- The book also contains numerous solved examples, illustrations, tables, diagrams, and exercises to enhance the understanding and practice of the subject.
- The book is available in paperback and e-book formats, and has been revised and updated in its 17th edition .

4. Engineering Chemistry by K. Sesha Maheswaramma, Pearson

- Engineering Chemistry is an interdisciplinary subject offered to undergraduate engineering students.
- The book introduces the fundamental concepts of chemistry in a simple and concise manner and highlights the role of chemistry in the field of engineering.
- The book covers topics such as water and its treatment, corrosion and its control, polymers and composites, fuels and combustion, nanomaterials and nanotechnology, and environmental chemistry.
- The book also provides numerous examples, solved problems, exercises, and multiple-choice questions to help students understand and apply the concepts.
- The book is designed to meet the syllabus requirements of various universities and technical boards.

5. Engineering Chemistry by OG Palanna, Mc Graw Hill Education, New Delhi

- This is a textbook that covers the fundamental concepts and applications of engineering chemistry for undergraduate engineering students.
- The book is divided into 16 chapters, each focusing on a specific topic such as water treatment, corrosion, polymers, fuels, electrochemistry, nanotechnology, etc.
- The book provides a balanced coverage of both theoretical and practical aspects of engineering chemistry, with numerous examples, problems, and case studies.
- The book also includes appendices, glossary, and index for easy reference and revision.
- The book is written in a clear and concise manner, with diagrams, tables, and figures to illustrate the concepts and data.
- The book is suitable for students of various branches of engineering, such as civil, mechanical, electrical, electronics, chemical, etc.

6. Engineering Chemistry by Shashi Chawala, Dhanpat Rai Publishing Comp, New Delhi

- Engineering Chemistry by Shashi Chawala is a textbook written for the first year engineering students of various branches in India.
- The book covers the basic concepts and applications of chemistry in engineering, such as water treatment, corrosion, polymers, fuels, lubricants, cement, glass, ceramics, etc.
- The book is divided into 26 chapters, each with a summary, review questions, and numerical problems.
- The book also contains experiments in applied chemistry, which can be performed in the laboratory.
- The book is widely recommended in most engineering colleges and universities in India, as it provides a comprehensive and systematic coverage of the syllabus.
- The book is written in a simple and lucid language, with diagrams, tables, and examples to illustrate the topics.
- The book is updated with the latest developments and trends in the field of engineering chemistry.

7. University Chemistry by BH Mahan

- University Chemistry by BH Mahan is a textbook of general chemistry for undergraduate students.
- The book covers the fundamental concepts and principles of chemistry, such as atomic structure, chemical bonding, thermodynamics, kinetics, equilibrium, electrochemistry, nuclear chemistry, organic chemistry, and biochemistry.

- The book also includes numerous examples, exercises, and problems to help students apply their knowledge and develop their problem-solving skills.
- The book is divided into four parts: Part I: The Structure of Matter, Part II: The Properties of Matter, Part III: Chemical Reactions, and Part IV: Chemistry of the Elements.
- The book is written in a clear and concise style, with an emphasis on the logical development of ideas and the use of mathematical tools.
- The book is suitable for students who have a basic background in mathematics and physics, and who are interested in learning the scientific basis of chemistry.

8. University Chemistry by CNR Rao

- University Chemistry by CNR Rao is a textbook for undergraduate students of chemistry, covering the basic concepts and principles of physical, inorganic and organic chemistry.
- The book is divided into three parts: Part I deals with the structure and bonding of atoms and molecules, thermodynamics, chemical kinetics, electrochemistry and surface chemistry. Part II covers the periodic properties of elements, coordination chemistry, nuclear chemistry and solid state chemistry. Part III introduces the fundamentals of organic chemistry, including hydrocarbons, functional groups, stereochemistry, reaction mechanisms and biomolecules.
- The book aims to provide a comprehensive and rigorous treatment of the subject, with numerous examples, exercises, tables and figures to illustrate the concepts and applications. The book also includes appendices on mathematical methods, units and constants, and a glossary of terms.
- The book is written by CNR Rao, a distinguished Indian chemist and professor, who has made significant contributions to the fields of solid state and materials chemistry, nanoscience and nanotechnology. He has authored over 1600 research papers and 50 books, and has received many national and international awards and honors, including the Bharat Ratna, the highest civilian award of India.